PFL-AI
Body of Knowledge

The reference for the PCI AI Project Finance Leader
Project economics · structuring · financial mathematics · the governed use of AI

FIRST EDITION — PHASE 1 PROTOTYPE (NOT FOR RELEASE)
PROJECT CONTROLS INSTITUTE GLOBAL

Finance intelligently. Control predictively. Deliver successfully.

AI proposes; the professional verifies, decides and remains accountable.

Prototype notice

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PART ONE

Foundations

Domains 1–4 — the profession, accounting foundations, financial mathematics and investment appraisal: the financial grammar of project finance leadership.

03

DOMAIN 3

Time Value of Money and Financial Mathematics (quantitative flagship)

Group: Foundations (Domain 3 of 4 in Part One). **Target:** ~72 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain is the definitive home of the discounting symbols — `PV(x)`, `FV(x)`, `DF(t)`, `r`, `n`, `i_nom`, `i_real`, `π`, annuity payment `A` — every later domain restates them from here. Because this book is discounting-heavy, `PV` written bare is reserved for Earned Value contexts only; present value is always written `PV(x)` or in words. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Why this domain exists

Every judgment a project finance leader makes — whether an investment is worth making (Domain 4), how much debt a project can carry (Domain 10), what a concession's payment stream is worth (Domain 7), whether a delay destroys value (Domain 8) — reduces at some point to one question: what is money at one date worth at another date? Time value of money (TVM) is the machinery that answers it. This domain builds that machinery from first principles: interest and growth (KA 3.1), the annuity family and loan schedules that debt structures are made of (KA 3.2), and the adjustments — inflation, escalation and currency — that separate a defensible model from a spreadsheet accident (KA 3.3). Nothing in this domain is optional equipment: the financial models of Domain 6 are, in the end, disciplined arrangements of exactly these calculations, and the coverage ratios lenders live by (Domain 10) are discounted cash flows wearing covenants.

Learning objectives. After this domain a candidate can: distinguish simple and compound interest and compute either; move any cash amount across time with `FV(x)` and `PV(x)` and read a discount-factor table; value level streams with annuity and perpetuity formulae; build and check a loan schedule (annuity, level-principal and bullet); convert between nominal, periodic and effective rates; keep nominal and real quantities consistent using the Fisher relation; escalate costs and revenues defensibly; state how currency and forward rates enter project cash flows; and apply the governed-AI rule to any machine-produced calculation of these kinds.

The master financing. One fictional project runs through this domain and returns in Domains 4, 6 and 10. **Kestrel Water SPC** is a special-purpose company developing a seawater desalination plant. Its senior lenders offer a **USD 42,000,000** loan repayable over **12 years at 6.0 %** annual interest; the offtaker will pay an **availability payment of USD 5,600,000 per year for 25 years**; Kestrel's board evaluates offers at a discount rate of **8.0 %**. All three numbers are used repeatedly below — by the end of the domain the reader can price every side of Kestrel's deal.

KNOWLEDGE AREA 3.1

Interest and value

Topics: 3.1.1 simple and compound interest · 3.1.2 present and future value · 3.1.3 discount factors.

3.1.1 Simple and compound interest

Definitions. Interest is the price of money over time. Under **simple interest** the charge accrues on the original principal only:

```
FV(x) = x × (1 + r × n)        — simple interest
```

Under **compound interest** each period's interest joins the principal and itself earns interest:

```
FV(x) = x × (1 + r)^n          — compound interest
```

where `x` is the amount today (currency), `r` the interest rate per period (ratio) and `n` the number of periods (count). Project finance is a compound-interest world: loans, discounting, escalation and returns all compound. Simple interest survives only in narrow corners — short-duration instruments, some penalty-interest clauses, and certain Islamic-finance structures that avoid interest altogether and price time differently (Domain 9, KA 9.3).

The principle. Compounding is growth on growth. Its effect looks negligible over one or two periods and decisive over ten: the gap between the two formulae widens geometrically, which is why long-life infrastructure — concessions of 25 to 40 years — is acutely sensitive to the rate used.

WORKED EXAMPLE

Worked example 3.1.1 — the same deposit, two rules.

1. **Setup.** USD 100,000 is placed for 3 years at 8.0 % per year. Compare simple and compound growth.
2. **Formula.** Simple: `FV(x) = x(1 + rn)`. Compound: `FV(x) = x(1 + r)^n`, with `x` = 100,000, `r` = 0.08, `n` = 3.
3. **Substitution.** Simple: `100,000 × (1 + 0.08 × 3) = 100,000 × 1.24`. Compound: $100{,}000 \times 1.08^3 = 100{,}000 \times 1.259712$.
4. **Result.** Simple: **USD 124,000**. Compound: **USD 125,971** (125,971.20 at full precision).
5. **Interpretation.** Three years of compounding adds USD 1,971 over simple interest — under 2 %. Over a 25-year concession life the same 8 % compounds to $1.08^{25} \approx 6.85$ times the principal, while simple interest reaches only 3.0 times. The professional habit this example teaches: never extrapolate a short-horizon intuition to a long-horizon contract.

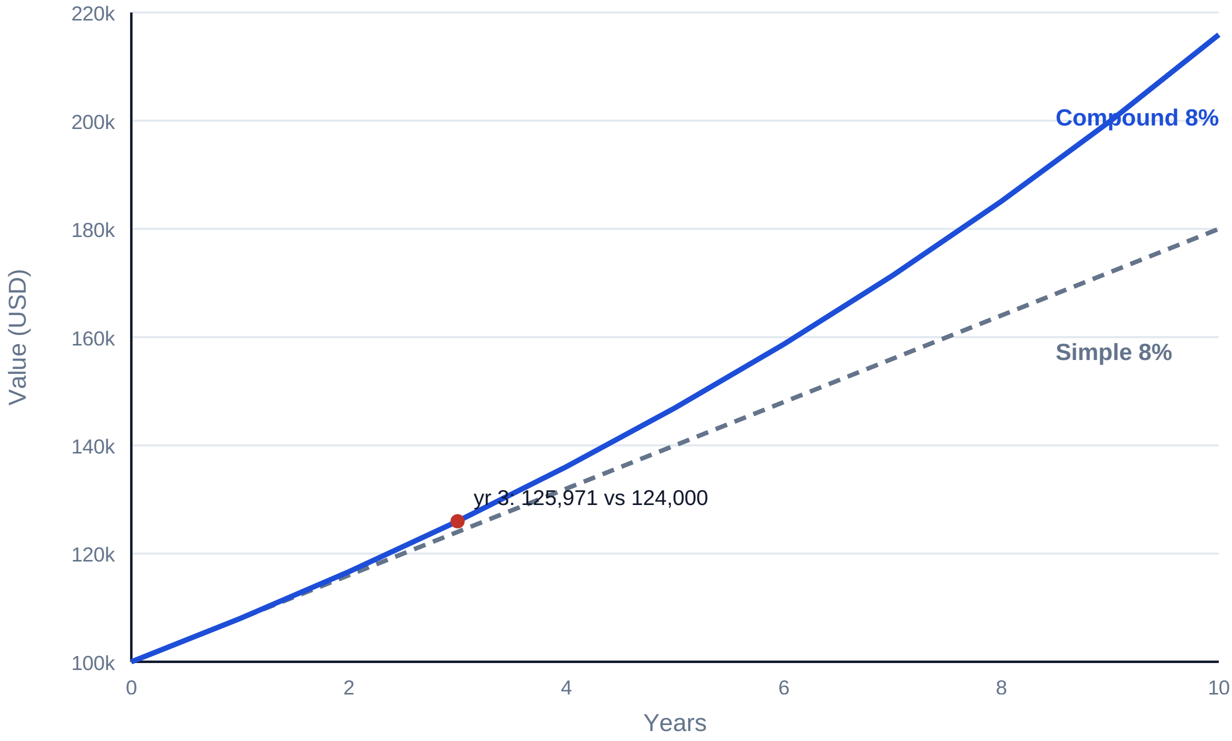

Fig 3.1.1 — Simple versus compound growth of USD 100,000 at 8 %

3.1.2 Present and future value

Definitions. Compounding runs both ways. **Future value** carries an amount forward; **present value** brings a future amount back:

```
FV(x) = x × (1 + r)^n          PV(x) = x / (1 + r)^n
```

`PV(x)` answers the only question a bid evaluation, a settlement offer or a buy-out negotiation really asks: what is that future money worth today, at the return I could otherwise earn? The rate `r` used in this role is called the **discount rate**; choosing it is a Domain 4 judgment (and, for a levered project, a Domain 9/10 conversation) — this KA takes it as given.

WORKED EXAMPLE

Worked example 3.1.2 — pricing a deferred receipt.

1. **Setup.** Kestrel will receive a USD 500,000 connection rebate exactly 5 years from now. The board discounts at 7.0 % for this risk. What is the rebate worth today?
2. **Formula.** `PV(x) = x / (1 + r)^n`, with `x` = 500,000, `r` = 0.07, `n` = 5.
3. **Substitution.** `PV(x) = 500,000 / 1.07⁵ = 500,000 / 1.402552`.
4. **Result. USD 356,493** (356,493.09 at full precision; indicatively ≈ SAR 1,336,849 at `USD 1 ≈ SAR 3.75`).
5. **Interpretation.** A five-year wait at 7 % costs the rebate just under 29 % of its face value. If a counterparty offered USD 380,000 cash today to extinguish the obligation, Kestrel should accept — and if offered USD 330,000, decline. `PV(x)` turns "later" into a number that can be compared, which is the whole trade of this profession.

Common pitfall. Discounting with simple interest — `500,000 / (1 + 0.07 × 5) = 370,370` — overstates the value by USD 13,877 here. The error grows with horizon and rate; audit any model whose discount factors were "simplified".

3.1.3 Discount factors

Definition. The **discount factor** for period `t` is the present value of one currency unit received at `t`:

```
DF(t) = 1 / (1 + r)^t          so that   PV(x at t) = x × DF(t)
```

Discount factors are how models industrialise discounting: compute the factor row once, multiply every cash-flow row by it (Domain 6, KA 6.1). They also make review fast — an experienced reviewer reads a factor table the way a scheduler reads float.

At `r` = 10 %:

`t` (years)	1	2	3	4	5
`DF(t)`	0.9091	0.8264	0.7513	0.6830	0.6209

Two sanity rules a reviewer applies on sight: factors must **decline monotonically**, and each factor must equal the previous factor divided by `(1 + r)`. A factor table that violates either has a hard-coded cell or a broken formula — a Domain 6 model-check classic (KA 6.4).

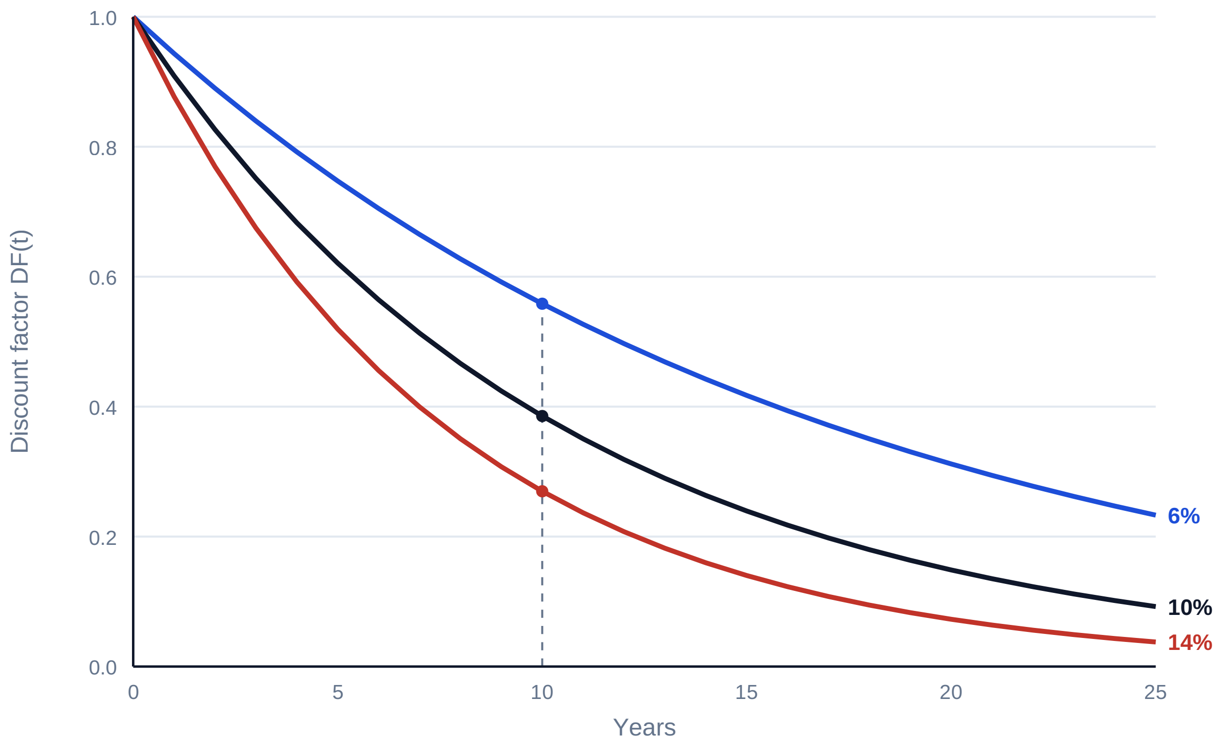

Fig 3.1.2 — Discount-factor curves at 6 %, 10 % and 14 %

Rounding discipline. Factors are displayed to four decimals but **calculations use full precision**. Multiplying a USD 900,000,000 programme cash flow by a four-decimal factor can move results by tens of thousands; the display is for the reader, never for the arithmetic (see the registry's decimal-arithmetic rule).

AI in this KA

Spreadsheet copilots will happily draft TVM formulae, and large-language-model assistants will "explain" a discount factor with perfect fluency and an inverted exponent. The governed position (the PCI principle: **AI proposes; the professional verifies, decides and remains accountable**) is mechanical here because verification is cheap: recompute one factor by hand, check monotonicity, check `DF(1) × (1+r) = 1`. An AI-drafted factor row that no human recomputed is not a calculation; it is a claim.

KEY TERMS — KA 3.1

Term	Meaning
Simple interest	Interest accruing on original principal only: `FV(x) = x(1 + rn)`.
Compound interest	Interest accruing on principal and accumulated interest: `FV(x) = x(1 + r)^n`.
Present value `PV(x)`	Today's worth of a future amount at discount rate `r`.
Future value `FV(x)`	The compounded worth of an amount at a future date.
Discount rate	The rate `r` used to move value across time; the required return for the risk.
Discount factor `DF(t)`	`1/(1+r)^t`; the PV of one unit received at `t`.

SAMPLE MCQS — KA 3.1

MCQ 3.1-A `[3.1.2 · Application]` A payment of USD 500,000 is due in 5 years; the discount rate is 7 %. Its present value is closest to: - A. USD 370,370 - B. USD 356,493 - C. USD 381,448 - D. USD 701,276

Rationale: $500{,}000 / 1.07^5 = 356{,}493$. A discounts with simple interest `(1 + 0.35)`; C uses four years instead of five; D compounds forward instead of discounting back.

MCQ 3.1-B `[3.1.3 · Application]` At a 10 % discount rate, the discount factor for year 3 is: - A. 0.7000 - B. 0.8264 - C. 0.7513 - D. 0.6830

Rationale: $1/1.10^3 = 0.7513$. A subtracts 10 % three times (simple); B is the year-2 factor; D is the year-4 factor — the two most common off-by-one-period errors.

MCQ 3.1-C `[3.1.1 · Analysis]` A lender quotes 25-year money and a borrower's analyst tests affordability using simple interest "as a conservative shortcut". The analyst's error is that: - A. simple interest overstates the debt cost, so the test is merely too strict - B. compound growth exceeds simple growth over long horizons, so the shortcut materially understates the true accumulation - C. the two methods converge over long horizons, so the shortcut is harmless - D. simple interest cannot be computed for horizons beyond ten years

Rationale: Compounding produces growth on growth: at 8 % over 25 years the compound multiple is ≈ 6.85 versus 3.0 simple — the shortcut is anti-conservative, the opposite of the analyst's claim (so A is wrong); the methods diverge rather than converge (C); D is arithmetic nonsense.

MCQ 3.1-F `[3.1.2 · Application]` USD 250,000 is invested at 7 % compound for 6 years. Its future value is closest to: - A. USD 355,000 - B. USD 350,638 - C. USD 375,183 - D. USD 166,586

Rationale: $250{,}000 \times 1.07^6 = 375{,}183$. A is simple interest `(1 + 0.07 × 6)`; B compounds only 5 years; D divides instead of multiplying — discounting when the question asks for growth.

MCQ 3.1-D `[3.1.3 · Recall]` Which statement about discount factors is an invariant a reviewer can test without knowing the project? - A. `DF(t)` rises when cash flows are contracted - B. `DF(t+1) = DF(t) / (1 + r)` for every `t` - C. `DF(t)` equals `1 − r × t` - D. factors below 0.5 indicate an error

Rationale: Each factor is the prior factor discounted one more period — B holds for any `r`. A confuses risk of flows with the factor row; C is the simple-interest approximation; D is false — any long horizon at a normal rate passes 0.5 (see Fig 3.1.2).

MCQ 3.1-E `[3.1.2 · Application]` At 9 %, which is worth more today: USD 400,000 in 2 years, or USD 470,000 in 4 years? - A. the 470,000 — larger amounts always win - B. the 400,000: PV 336,672 vs 332,960 - C. the 470,000: PV 395,590 vs 336,672 - D. they are equal at 9 %

Rationale: $400{,}000/1.09^2 = 336{,}672$; $470{,}000/1.09^4 = 332{,}960$ — the earlier, smaller amount wins by USD 3,712. A ignores discounting entirely; C discounts the 470,000 across only two periods instead of four; D asserts an equality the arithmetic denies.

SELF-CHECK — KA 3.1

1. Why must discount factors decline monotonically? — Because `(1+r)^t` grows with `t` for any positive `r`; a rising factor implies a negative rate or a broken formula.
2. Your model shows *`DF(4) = 0.6830`* and *`DF(5) = 0.6209`* at 10 %. One cell check confirms both. Which? — `0.6830 / 1.10 = 0.6209`: each factor is the prior factor divided by `(1+r)`.
3. A colleague says "8 % for three years is 24 %". — Only under simple interest; compounded it is $1.08^3 - 1 = 25.97$ %, and the gap widens every further year.

KNOWLEDGE AREA 3.2

Annuities and loan schedules

Topics: 3.2.1 annuities and perpetuities · 3.2.2 loan schedules · 3.2.3 compounding frequency and effective rates.

3.2.1 Annuities and perpetuities

Definitions. An **annuity** is a level stream — the same amount each period for `n` periods. Project finance is built from annuities and near-annuities: availability payments, capacity charges, lease rentals, O&M fees, debt service. The present value of an ordinary annuity (payments at period-ends) of amount `A`:

```
PV(annuity) = A × AF(r, n)      where   AF(r, n) = (1 − (1 + r)^−n) / r
```

`AF(r, n)` is the **annuity factor** — the sum of the discount factors `DF(1) … DF(n)`. A **perpetuity** is the limiting case as `n → ∞`:

```
PV(perpetuity) = A / r
```

WORKED EXAMPLE

Worked example 3.2.1 — valuing Kestrel's availability stream.

1. **Setup.** Kestrel's offtaker will pay USD 3,000,000 per year for 20 years under an early tariff proposal, paid annually in arrears. The board discounts at 9.0 %. Value the stream.
2. **Formula.** `PV = A × AF(r, n)`, `AF(r, n) = (1 − (1+r)^−n)/r`, with `A` = 3,000,000, `r` = 0.09, `n` = 20.
3. **Substitution.** `AF(0.09, 20) = (1 − 1.09⁻²⁰)/0.09 = 9.128546`; `PV = 3,000,000 × 9.128546`.
4. **Result. USD 27,385,637** (using the full-precision factor).
5. **Interpretation.** Twenty years of payments are worth barely nine years' worth of face value — discounting at 9 % has priced away more than half the nominal USD 60,000,000 total. Sensitivity: at a 10 % rate the factor falls to 8.513564 and the value to USD 25,540,691 — a 6.7 % value loss for one point of rate, the shape of sensitivity every concession bidder lives with (Domain 7, KA 7.4).

Common pitfall — the rounded factor. Using a four-decimal factor (9.1285) gives USD 27,385,500 — USD 137 adrift here, but scale the stream by 100× and the drift is five figures. Factors are display; arithmetic is full precision (KA 3.1.3).

Timing variants. An **annuity-due** (payments in advance, common in leases and some availability regimes) is worth `(1 + r)` times the ordinary annuity; a **deferred annuity** starts after a gap and is discounted twice — as a stream, then back across the deferral. Misreading the payment convention in a concession agreement misprices the whole stream by one period's discounting — check the contract, not the habit (Domain 12, KA 12.2).

WORKED EXAMPLE

Worked example 3.2.1b — the lease quoted in advance.

1. **Setup.** Kestrel's operations base is leased at USD 500,000 per year for 5 years, payable **in advance**. Discount rate 8.0 %. Value the obligation.
2. **Formula.** `PV(due) = A × AF(r, n) × (1 + r)`, with `A` = 500,000, `r` = 0.08, `n` = 5.
3. **Substitution.** `AF(0.08, 5) = 3.992710`; ordinary value `500,000 × 3.992710 = 1,996,355`; due value `1,996,355 × 1.08`.
4. **Result. USD 2,156,063** (2,156,063.42). The in-advance convention adds USD 159,708 — exactly one period's discounting on the whole stream.
5. **Interpretation.** Eight per cent of the stream's value hangs on one word in the lease. Reading "payable annually in advance" as an ordinary annuity is the commonest annuity mistake in diligence reviews (Domain 13, KA 13.1) — and it always flatters the tenant's model.

WORKED EXAMPLE

Worked example 3.2.1c — the deferred availability stream.

1. **Setup.** A grantor variant offers Kestrel the same USD 5,600,000 × 25-year availability supplement, but with the **first payment at the end of year 4** (a three-year deferral while the plant ramps up). Discount rate 8.0 %.
2. **Formula.** `PV = A × AF(r, n) / (1 + r)^d`, `d` = 3 (the stream's own valuation point sits at the end of year 3, one period before its first payment).
3. **Substitution.** `5,600,000 × 10.674776 = 59,778,747` (the stream valued at end-year 3); `59,778,747 / 1.08³ = 59,778,747 / 1.259712`.
4. **Result. USD 47,454,296**.
5. **Interpretation.** The three-year wait costs USD 12.3 million — a fifth of the stream's value — which is precisely what the grantor saves by deferring. Any negotiation over payment timing is a negotiation over value at compound rates; Case study A prices the whole family of such trades.

3.2.2 Loan schedules

The three canonical shapes. A loan of principal `P` over `n` periods at rate `r` can be scheduled three ways, and a project finance leader must read all three on sight:

Shape	Rule	Debt-service profile	Where seen
Annuity (equal instalment)	Payment `A = P × r / (1 − (1+r)^−n)` each period	Level total; interest share falls, principal share rises	Most project term loans; mortgages
Level principal	Principal `P/n` each period + interest on balance	Front-loaded total, declining	Some ECA and development-bank facilities (Domain 9)
Bullet	Interest only; principal entire at maturity	Low until the final balloon	Bonds; mini-perm structures; refinancing plays (Domain 15)

WORKED EXAMPLE

Worked example 3.2.2 — Kestrel's senior loan.

1. **Setup.** USD 42,000,000, 12 annual instalments, 6.0 % — annuity shape. Find the instalment and build the first three schedule rows.
2. **Formula.** `A = P × r / (1 − (1+r)^−n)`, with `P` = 42,000,000, `r` = 0.06, `n` = 12. Each row: interest = opening balance × `r`; principal = `A` − interest; closing = opening − principal.
3. **Substitution.** `A = 42,000,000 × 0.06 / (1 − 1.06⁻¹²) = 42,000,000 / 8.383844`.
4. **Result.** `A` = **USD 5,009,635** per year (5,009,635.23; indicatively ≈ SAR 18,786,132). Schedule:

Year	Opening balance	Interest (6 %)	Principal	Closing balance
1	42,000,000	2,520,000	2,489,635	39,510,365
2	39,510,365	2,370,622	2,639,013	36,871,351
3	36,871,351	2,212,281	2,797,354	34,073,997

1. **Interpretation.** The instalment never changes; its composition does — year 1 is 50 % interest, and by the final year almost all principal. Two checks a reviewer runs instantly: the closing balance after year 12 must be exactly zero (a residual means a formula error), and total principal across all rows must equal `P`. This schedule is the direct input to Kestrel's debt service line — and therefore to its DSCR — in Domains 6 and 10.

WORKED EXAMPLE

Worked example 3.2.2b — the same loan, level-principal.

1. **Setup.** Kestrel's ECA co-lender offers the same USD 42,000,000 over 12 years at 6.0 % but on a **level-principal** schedule. Build the first two rows and the final row, and compare lifetime interest across all three shapes.
2. **Formula.** Principal each year `P/n = 42,000,000/12 = 3,500,000`; interest = opening balance × 6 %; total = principal + interest.
3. **Substitution.** Year 1: `3,500,000 + 42,000,000 × 0.06`. Year 2: `3,500,000 + 38,500,000 × 0.06`. Year 12: `3,500,000 + 3,500,000 × 0.06`.
4. **Result.** Year 1 **USD 6,020,000** · year 2 **USD 5,810,000** · year 12 **USD 3,710,000** — a declining profile. Lifetime interest: level-principal **USD 16,380,000** · annuity **USD 18,115,623** · bullet **USD 30,240,000**.
5. **Interpretation.** Level-principal is the cheapest shape in total interest because principal retires fastest — but its year-1 debt service is USD 1,010,365 heavier than the annuity's, exactly when a ramping project is cash-poorest. Shape selection is a cash-timing decision, not a cost minimisation: Domain 10 (KA 10.1) sizes debt against the early-year coverage this choice determines.

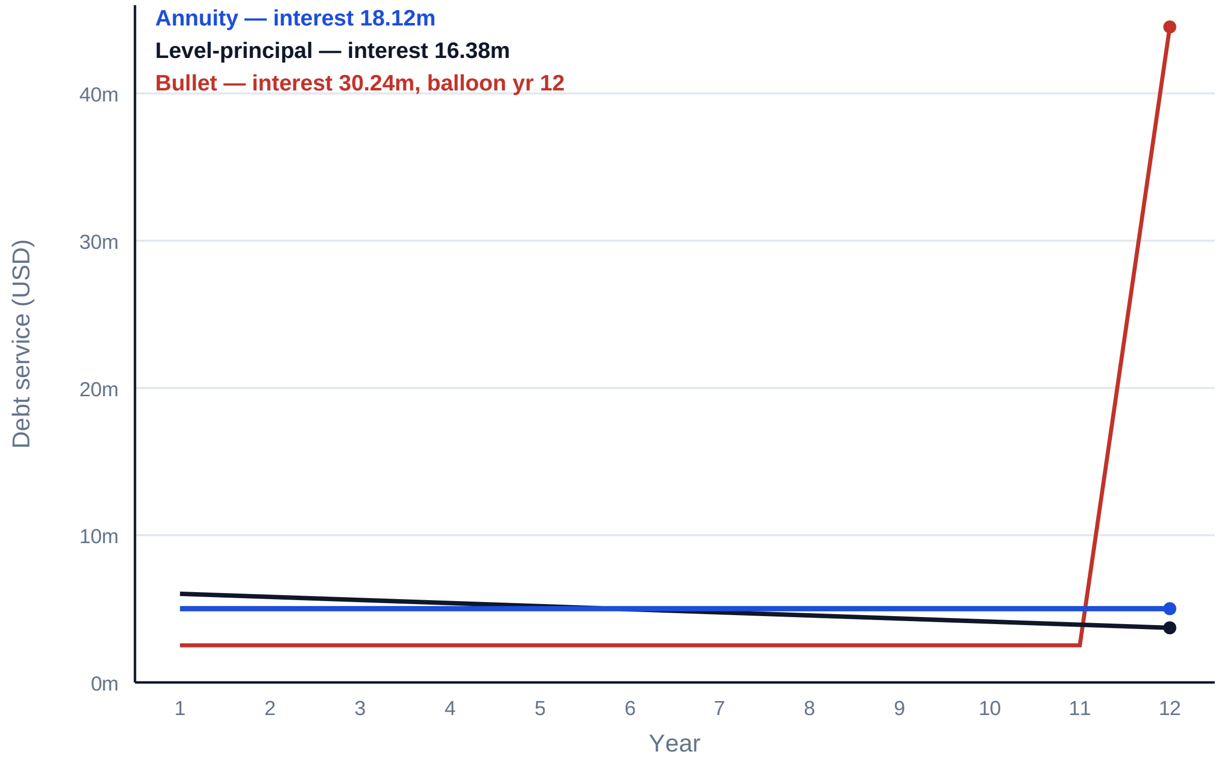

Fig 3.2.2 — Three shapes, one loan: annual debt service under annuity, level-principal and

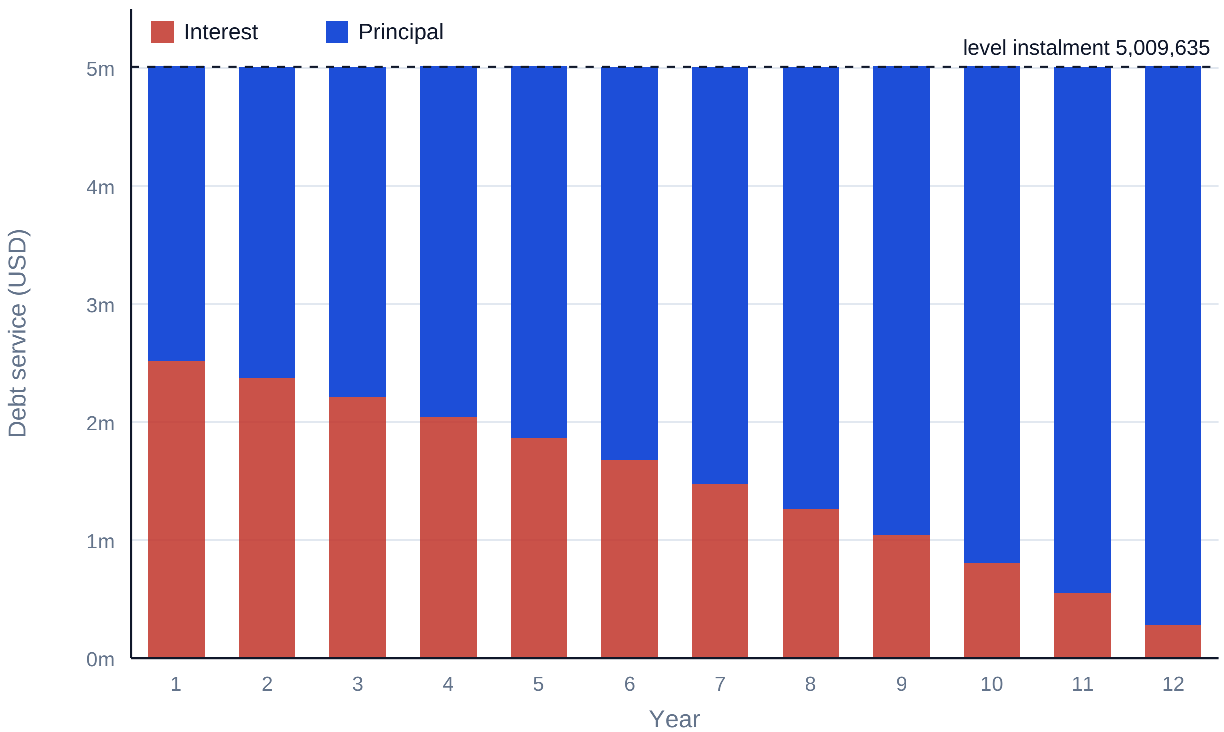

Fig 3.2.1 — Anatomy of an annuity loan: Kestrel's USD 42,000,000, 12 years at 6 %

Common pitfall — the bullet that looks cheap. A bullet's early debt service is a fraction of an annuity's, and a cash-strapped sponsor will feel the temptation. The full price appears at maturity as refinancing risk — the balloon must be refinanced at whatever the market then charges (Domain 15, KA 15.3). Case study B prices this trade.

3.2.3 Compounding frequency and effective rates

Definitions. A quoted **nominal annual rate** `i_nom` compounded `m` times per year applies `i_nom/m` per period. The **effective annual rate (EAR)** is the once-a-year rate that produces the same growth:

```
EAR = (1 + i_nom / m)^m − 1
```

WORKED EXAMPLE

Worked example 3.2.3 — the same 6 %, four ways.

1. **Setup.** A facility is quoted at "6 % nominal". Find the effective annual rate if compounding is annual, semi-annual, quarterly and monthly.
2. **Formula.** `EAR = (1 + 0.06/m)^m − 1` for `m` = 1, 2, 4, 12.
3. **Substitution.** `m=4`: `(1 + 0.015)⁴ − 1 = 1.061364 − 1`.
4. **Result.** Annual **6.000 %** · semi-annual **6.090 %** · quarterly **6.136 %** · monthly **6.168 %**.
5. **Interpretation.** "Six per cent" is not one price; it is four prices in this example alone, and the differences compound over a 12-year loan. Term sheets are compared on effective rates, never on quoted nominals — and loan models must use the **periodic** rate `i_nom/m` with `n × m` periods, not the EAR with `n` years, or debt service will be misstated (Domain 6, KA 6.3).

AI in this KA

Amortisation is where AI-assisted modelling earns its keep — and where it must be caged. Generating a 360-row monthly schedule is exactly the mechanical work a copilot does well; whether the schedule uses the periodic rate correctly, honours the day-count convention in the actual facility agreement, and zeroes the final balance is exactly what it gets wrong silently. The governed workflow (Domain 6, KA 6.4; Domain 16): AI drafts the schedule; the professional runs the three deterministic checks (final balance zero; Σ principal = `P`; payment recomputed independently) before any output leaves the model.

KEY TERMS — KA 3.2

Term	Meaning
Annuity / annuity-due	Level periodic stream, in arrears / in advance.
Annuity factor `AF(r,n)`	`(1 − (1+r)^−n)/r`; PV of 1 per period for `n` periods.
Perpetuity	Level stream forever; `PV = A/r`.
Annuity, level-principal, bullet	The three canonical loan shapes.
Balloon / refinancing risk	The bullet's maturity principal and the risk of the rate then prevailing.
Nominal rate / EAR	Quoted annual rate with compounding frequency / its once-a-year equivalent.

SAMPLE MCQS — KA 3.2

MCQ 3.2-A `[3.2.2 · Application]` A USD 42,000,000 loan is repaid by 12 equal annual instalments at 6 %. The instalment is closest to: - A. USD 3,500,000 - B. USD 2,520,000 - C. USD 5,009,635 - D. USD 5,706,454

Rationale: $42{,}000{,}000 \times 0.06/(1 - 1.06^{-12}) = 5{,}009{,}635$. A divides principal by 12 and ignores interest; B is interest-only on the full balance; D uses a 10-year annuity factor — the wrong-tenor error.

MCQ 3.2-B `[3.2.3 · Application]` A 6 % nominal rate compounded quarterly has an effective annual rate of: - A. 6.00 % - B. 6.09 % - C. 6.14 % - D. 6.17 %

Rationale: $(1 + 0.015)^4 - 1 = 6.136\ \% \approx 6.14\ \%$. A is the nominal itself; B is semi-annual compounding; D is monthly — each a frequency misread.

MCQ 3.2-C `[3.2.1 · Analysis]` Two otherwise identical concession bids value the same 20-year availability stream. Bid X used the full-precision annuity factor; Bid Y rounded the factor to four decimals. The most defensible statement is: - A. the bids differ by an amount that grows with the size of the stream, and only X's practice passes model audit - B. rounding the factor is conservative, so Y understates value and is safer - C. the difference is always immaterial because four decimals is industry standard - D. Y's approach is wrong only if the discount rate exceeds 10 %

Rationale: Factor rounding error scales linearly with the cash flows and has no consistent conservative direction (B, D wrong); materiality depends on scale, not convention (C wrong); model audit standards require full-precision arithmetic with display-only rounding.

MCQ 3.2-F `[3.2.1 · Analysis]` A wayleave pays USD 800,000 per year; the discount rate is 8 %. Modelling it as a 30-year annuity instead of a perpetuity understates its value by: - A. USD 993,773 — the present value of the post-year-30 tail - B. nothing material — thirty years is effectively forever - C. USD 4,000,000 — one-third of the perpetuity value - D. it overstates value, because perpetuities are riskier

Rationale: Perpetuity $800{,}000/0.08 = 10{,}000{,}000$; 30-year annuity $800{,}000 \times 11.257783 = 9{,}006{,}227$; the gap — 9.9 % of value — is the discounted tail. B waves away a million dollars; C invents a fraction; D confuses valuation arithmetic with a risk adjustment that belongs in the rate, not the formula choice.

MCQ 3.2-D `[3.2.1 · Application]` A 5-year, USD 500,000-per-year lease payable **in advance** is valued at 8 %. Its present value is closest to: - A. USD 1,996,355 - B. USD 2,156,063 - C. USD 2,500,000 - D. USD 1,848,477

Rationale: Annuity-due = ordinary annuity × $(1+r)$: $500{,}000 \times 3.992710 \times 1.08 = 2{,}156{,}063$. A prices it as an ordinary annuity (the classic misread); C is the undiscounted total; D divides the ordinary value by 1.08 — the adjustment applied backwards.

MCQ 3.2-E `[3.2.2 · Analysis]` For the same principal, tenor and rate, which repayment shape has the lowest lifetime interest, and why? - A. the annuity — its instalments are level, so interest is averaged down - B. the bullet — deferral shrinks the money's time exposure - C. level-principal — the balance falls fastest, so less principal is outstanding for less time - D. all three are equal because rate and tenor are equal

Rationale: Interest is rate × outstanding balance × time; level-principal retires balance fastest (Kestrel: 16.38m vs 18.12m annuity vs 30.24m bullet). A's "averaging" is not a mechanism; B reverses the truth — the bullet maximises exposure; D ignores the balance path entirely.

SELF-CHECK — KA 3.2

1. What two instant checks validate any amortising schedule? — Final closing balance exactly zero; total principal equals the original loan.
2. Why is an annuity-due worth $(1+r)$ × the ordinary annuity? — Every payment arrives one period earlier, so each escapes one period of discounting.
3. A term sheet quotes "5.9 % monthly"; another "6.0 % annual". Which is dearer? — The first: $(1 + 0.059/12)^{12} - 1 = 6.06\ \%$ effective, above 6.00 %.

KNOWLEDGE AREA 3.3

Real-world adjustments: inflation, escalation and currency

Topics: 3.3.1 nominal and real rates · 3.3.2 inflation and escalation · 3.3.3 currency effects · 3.3.4 day-count conventions.

3.3.1 Nominal and real rates — the Fisher relation

Definitions. A **nominal** rate i_nom is quoted in money terms; a **real** rate i_real is measured in purchasing power, after inflation π. They are linked by the **Fisher relation**:

$$(1 + i\_nom) = (1 + i\_real) \times (1 + \pi) \qquad \text{so} \qquad i\_real = (1 + i\_nom)/(1 + \pi) - 1$$

The subtraction shortcut $i\_real \approx i\_nom - \pi$ is a first-order approximation only — usable for intuition, never for models.

WORKED EXAMPLE

Worked example 3.3.1 — the real cost of Kestrel's equity hurdle.

1. **Setup.** Kestrel's shareholders require a 9.0 % nominal return; inflation is expected at 3.0 %. What real return are they actually demanding?
2. **Formula.** $i\_real = (1 + i\_nom)/(1 + \pi) - 1$, with i_nom = 0.09, π = 0.03.
3. **Substitution.** $i\_real = 1.09 / 1.03 - 1 = 1.058252 - 1$.
4. **Result. 5.83 %** real (5.8252 % at full precision). The subtraction shortcut says 6.00 % — 17 basis points high.
5. **Interpretation.** Seventeen basis points sounds academic until it is applied to a 25-year stream: mispricing the hurdle by that much moves a large concession's value by roughly one and a half years' worth of payments. Models must live entirely in one world — nominal cash flows with nominal rates, or real with real. **Mixing them is the single most common TVM defect found in model audits** (Domain 6, KA 6.4; Domain 13, KA 13.2).

3.3.2 Inflation and escalation

Definitions. Inflation is the general price level's drift; **escalation** is the contractual or forecast growth of a specific price — a tariff indexed to CPI, labour rates under a collective agreement, steel under a commodity formula. An amount x escalating at rate e for n periods becomes $x \times (1 + e)^n$. Escalation compounds, exactly like interest — and like interest, it is routinely underestimated over long horizons.

WORKED EXAMPLE

Worked example 3.3.2 — an O&M contract's third-year price.

1. **Setup.** Kestrel's O&M contract prices year-0 services at USD 10,000,000, escalating 4.0 % annually. What is the year-3 price?
2. **Formula.** $x_n = x_0 \times (1 + e)^n$, with x_0 = 10,000,000, e = 0.04, n = 3.
3. **Substitution.** $10{,}000{,}000 \times 1.04^3 = 10{,}000{,}000 \times 1.124864$.
4. **Result. USD 11,248,640**.
5. **Interpretation.** Simple-escalation thinking says "+12 %" (USD 11,200,000); compounding adds USD 48,640 by year 3 and the gap widens every year after. In a 25-year operating model the compound-vs-simple escalation choice changes lifecycle cost by whole percentage points — Domain 8 (KA 8.2) builds the full escalated cash-flow machinery on this rule.

Indexation discipline. Real contracts escalate on published indices with definitions, lags, caps and floors (Domain 7, KA 7.3; Domain 12). A model's escalation assumptions must cite the contractual index and its mechanics, not a bare percentage — the toolkit's escalation register (3.T.3) exists for exactly this.

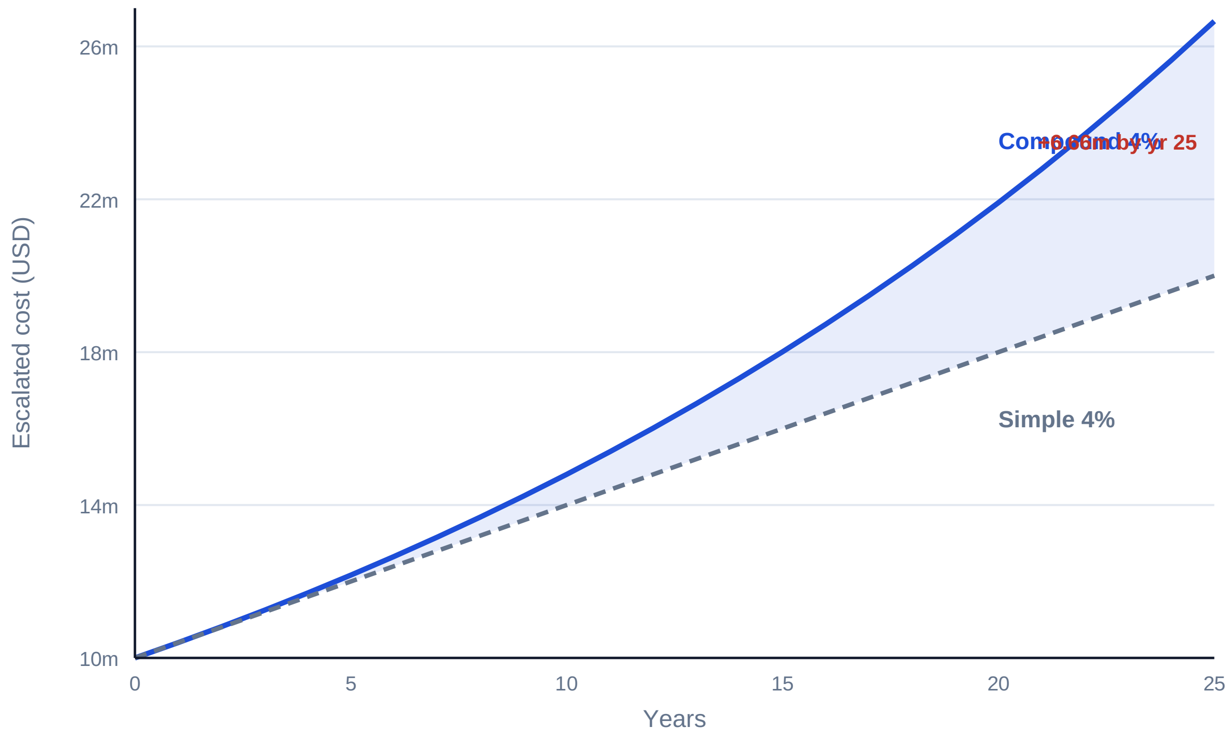

Fig 3.3.2 — Compound versus simple escalation of a USD 10,000,000 cost at 4 %

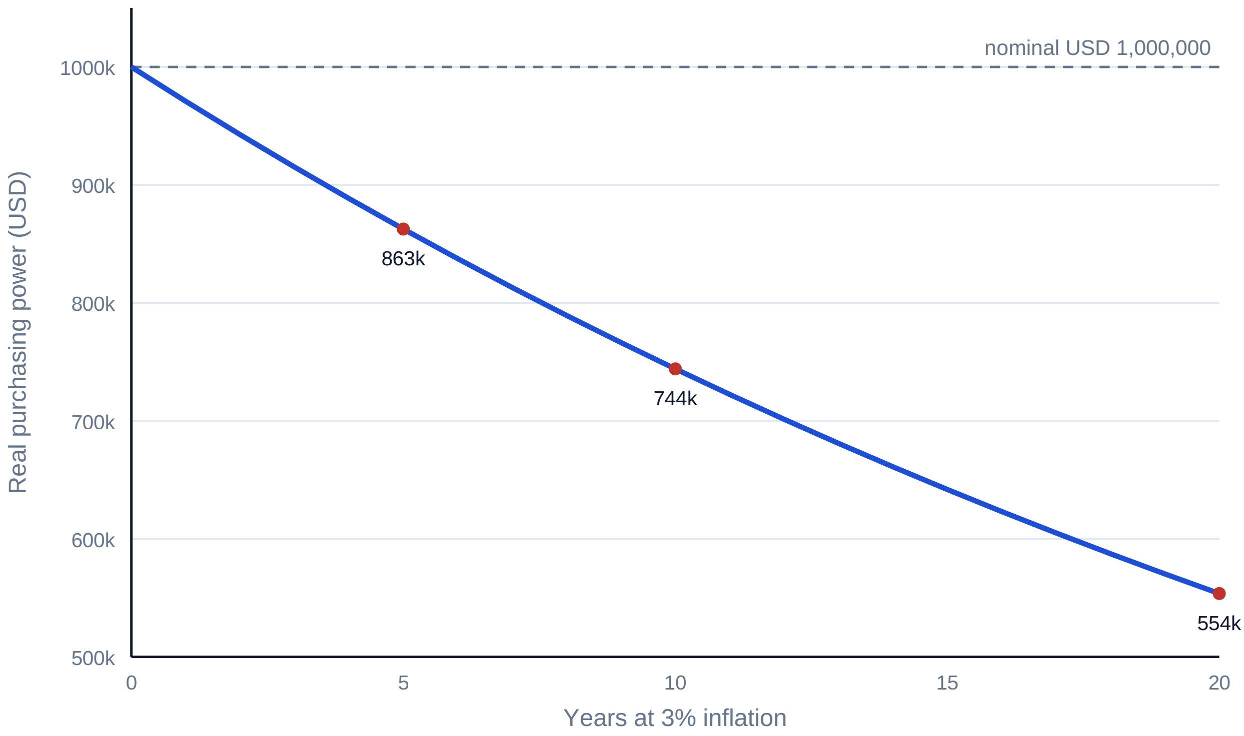

Fig 3.3.1 — What 3 % inflation does to USD 1,000,000 of purchasing power

3.3.3 Currency effects

The problem. Project cash flows often arrive in one currency while debt service, equipment or returns are owed in another. Two rates matter: the **spot rate** now, and the **forward rate** at which future exchange can be contracted. Forwards are priced off interest differentials (covered interest parity, stated here in its textbook form):

```
F ≈ S × (1 + i_quote) / (1 + i_base)
```

WORKED EXAMPLE

Worked example 3.3.3 — a one-year SAR/USD forward.

1. **Setup.** Spot `USD 1 = SAR 3.7500` (the indicative peg rate used throughout this book). One-year money costs 5.0 % in USD and 5.5 % in SAR. Estimate the one-year forward.
2. **Formula.** `F = S × (1 + i_SAR)/(1 + i_USD)`, with `S` = 3.7500.
3. **Substitution.** `F = 3.7500 × 1.055 / 1.05`.
4. **Result. SAR 3.7679 per USD**.
5. **Interpretation.** The forward is not a forecast; it is arithmetic — the rate at which borrowing in one currency and lending in the other breaks even. A project that leaves a currency mismatch unhedged is not "expecting the spot to hold"; it is running an open position whose size Domain 11 (KA 11.3) teaches you to measure and whose contractual mitigations Domain 12 allocates. **Assumption honesty:** actual pegs, spreads and convertibility restrictions are jurisdiction- and time-specific; every currency figure in this book is illustrative.

3.3.4 Day-count conventions

The problem. "Three months' interest" is not one number. Facility agreements specify a **day-count convention** — how days are counted and over what year they are divided — and the convention moves real money on large balances:

Convention	Rule	Typical home
30/360	Every month 30 days, year 360	Bonds, some term loans
actual/360	Actual days elapsed, year 360	Money markets, most floating-rate loans
actual/365	Actual days elapsed, year 365	Sterling markets, some jurisdictions

WORKED EXAMPLE

Worked example 3.3.4 — one quarter, three conventions.

1. **Setup.** Interest accrues on Kestrel's USD 42,000,000 drawn balance at 6.0 % for one calendar quarter that contains **92 actual days**. Compute the interest under each convention.
2. **Formula.** Interest = balance × rate × (days counted / year basis).
3. **Substitution.** 30/360: `42,000,000 × 0.06 × 90/360`. actual/360: `× 92/360`. actual/365: `× 92/365`.
4. **Result.** 30/360 **USD 630,000** · actual/365 **USD 635,178** · actual/360 **USD 644,000**.
5. **Interpretation.** The spread between conventions is USD 14,000 on one quarter of one drawdown — and actual/360 is systematically the most expensive because it counts every real day over a short year. Models must implement the convention named in the facility agreement (Domain 12), and the assumption register (3.T.1) records it per instrument; "6 % divided by 4" is not a convention, it is a guess.

AI in this domain — the systematic view

TVM is deterministic, which makes it the safest place in finance to use machine assistance and the most dangerous place to trust it unchecked: a wrong exponent produces a plausible number, not an absurd one. The domain's governed workflow, applied wherever Domains 4–16 discount anything:

1. **AI proposes** — drafts the factor table, schedule or escalated series.
2. **Deterministic checks** — monotone factors; `DF(1)(1+r) = 1`; schedule zeroes; Σ principal = `P`; one hand-recomputed cell per block.
3. **Assumption register** — rate, basis (nominal/real), compounding frequency, index, and currency of every stream written down (toolkit 3.T.1).
4. **The professional decides and remains accountable** — no AI-produced number reaches a board paper, bid or lender report without a named human owner (Domain 16, KA 16.4).

KEY TERMS — KA 3.3

Term	Meaning
Nominal / real rate	Money-terms rate / purchasing-power rate; linked by Fisher.
Fisher relation	`(1+i_nom) = (1+i_real)(1+π)`.
Inflation `π` **/ escalation** `e`	General price drift / specific contractual price growth.
Indexation	Contractual escalation by a published index, with lags, caps, floors.
Spot / forward rate	Exchange rate now / contracted for a future date.
Covered interest parity	Forward ≈ spot × interest-ratio; the no-arbitrage forward.

SAMPLE MCQS — KA 3.3

MCQ 3.3-A `[3.3.1 · Application]` Nominal return 9 %, inflation 3 %. The real return is closest to: - A. 6.00 % - B. 5.83 % - C. 12.27 % - D. 3.00 %

Rationale: `1.09/1.03 − 1 = 5.83 %`. A is the subtraction approximation; C multiplies `1.09 × 1.03 − 1` (compounding inflation on instead of off); D confuses the inflation rate itself with the real return.

MCQ 3.3-B `[3.3.2 · Application]` A USD 10,000,000 cost escalating at 4 % per year is, in year 3: - A. USD 11,200,000 - B. USD 10,400,000 - C. USD 12,486,400 - D. USD 11,248,640

Rationale: $10,000,000 \times 1.04^3 = 11,248,640$. A escalates simply (3 × 4 %); B stops at one year; C applies four periods' escalation with a decimal slip.

MCQ 3.3-C `[3.3.1 · Analysis]` A model discounts real (uninflated) cash flows at the sponsor's nominal 9 % hurdle. The result: - A. overstates value, because inflation is counted twice - B. understates value, because inflation is removed from the flows but still charged in the rate - C. is correct if inflation is below 5 % - D. is correct because discount rates are always nominal

Rationale: Real flows with a nominal rate deduct inflation twice — once from the flows, once inside the rate — so value is systematically understated. The consistency rule admits no threshold (C) and no default (D); A reverses the direction of the error.

MCQ 3.3-F `[3.3.1 · Application]` A model shows a nominal cash flow of USD 2,000,000 in year 5; inflation is 3 %. Its value in today's purchasing power (real terms) is: - A. USD 2,000,000 — real and nominal are equal at year 5 - B. USD 1,725,218 - C. USD 1,700,000 - D. USD 2,318,548

Rationale: $2,000,000 / 1.03^5 = 1,725,218$ — deflating by the price level, not discounting for time value (that happens separately, at a real rate). A ignores five years of inflation; C deflates simply (`2,000,000 × (1 − 0.03 × 5) = 1,700,000`); D multiplies by 1.03^5 — inflating instead of deflating.

MCQ 3.3-D `[3.3.4 · Application]` A USD 42,000,000 balance accrues at 6 % over a 92-day quarter. Under **actual/360** the interest is: - A. USD 630,000 - B. USD 635,178 - C. USD 644,000 - D. USD 620,548

Rationale: `42,000,000 × 0.06 × 92/360 = 644,000`. A is 30/360 (90/360); B is actual/365; D uses 90/365 — mixing the two conventions' halves.

MCQ 3.3-E `[3.3.2 · Analysis]` A 25-year operating model escalates O&M at "4 % simple" to be "conservative". The effect is: - A. conservative — simple escalation always overstates cost - B. anti-conservative — compound escalation exceeds simple by an amount that grows every year, so late-life costs are materially understated - C. neutral — the choice only redistributes cost between years - D. correct — operating contracts always escalate simply

Rationale: Escalation compounds (contracts index on last year's indexed price): by year 25 the compound multiple $1.04^{25} \approx 2.67$ far exceeds the simple 2.00. A claims the reverse; C ignores the widening gap; D asserts a contractual universal that KA 3.3.2's indexation discipline contradicts.

SELF-CHECK — KA 3.3

1. State the consistency rule. — Nominal flows with nominal rates, real flows with real rates; never mixed in one calculation.
2. Why does a 25-year concession make the Fisher shortcut dangerous? — The 17 bp error (at 9 %/3 %) compounds across every one of 25 discount factors.
3. Is a forward rate a forecast? — No: it is the no-arbitrage consequence of today's spot and the two interest rates.

Advanced topics — Domain 3

3.A.1 Continuous compounding

As compounding frequency $m \to \infty$, $(1 + r/m)^{mn} \to e^{rn}$. USD 100,000 at 8 % for 3 years compounds continuously to $100{,}000 \times e^{0.24}$ = **USD 127,125** — against 125,971 annually. In project work continuous compounding appears mainly in derivative pricing and some academic appraisal literature; its practical value here is as a bound: no compounding frequency can push growth beyond e^{rn}.

3.A.2 Irregular timing and period conventions

Real cash flows do not arrive on anniversary dates. Three conventions bridge the gap: **exact-date discounting** (each flow discounted by its actual day count — the XNPV approach, standard in transaction models); the **mid-period convention** (flows treated as arriving mid-period, appropriate for continuous operations revenue); and **period-end** (conservative for receipts, default in lender base cases). The convention is an assumption — it belongs in the assumption register, it changes results by up to half a period's discounting, and model audits (Domain 13, KA 13.2) check that one convention is applied everywhere.

WORKED EXAMPLE

Worked example 3.A.2 — the payment on day 500.

1. **Setup.** A USD 1,000,000 completion payment falls on **day 500** from the valuation date; the discount rate is 9.0 % annual. Compare exact-date discounting with the period-end habit of "call it year 2".
2. **Formula.** Exact-date: $PV = x / (1 + r)^{(days/365)}$.
3. **Substitution.** $1{,}000{,}000 / 1.09^{(500/365)} = 1{,}000{,}000 / 1.09^{1.36986}$.
4. **Result. USD 888,650**. "Year 2" rounding gives 841,680 — **USD 46,970 low**; "year 1" gives 917,431 — USD 28,781 high.
5. **Interpretation.** Bucketing a mid-period flow to the nearer anniversary mis-states value by up to half a period's discounting — nearly 5 % here. Transaction models discount on dates; period models declare mid-period or period-end explicitly and apply it everywhere. The convention chosen matters less than its disclosure and consistency.

3.A.3 Term structures — when one rate is not enough

A single flat r assumes money has the same price at every horizon. Markets disagree: rates form a **term structure**, and precise valuation discounts each period's flow at that period's rate (equivalently, its own $DF(t)$ from a curve). Project models typically justify a flat rate as a deliberate simplification of a curve — acceptable when documented and stress-tested (Domain 6, KA 6.4), a silent error when inherited unexamined from a template.

WORKED EXAMPLE

Worked example 3.A.3 — flat rate versus the curve.

1. **Setup.** A contractor is owed USD 1,000,000 at the end of each of years 1 and 2. The spot curve prices 1-year money at 5.0 % and 2-year money at 7.0 %; the model uses a flat 6.0 %. How far wrong is the flat rate?
2. **Formula.** Curve: $PV = x/(1 + r_1) + x/(1 + r_2)^2$. Flat: $PV = x/(1+r) + x/(1+r)^2$.
3. **Substitution.** Curve: $1{,}000{,}000/1.05 + 1{,}000{,}000/1.07^2 = 952{,}381 + 873{,}439$. Flat: $943{,}396 + 889{,}996$.
4. **Result.** Curve **USD 1,825,820**; flat **USD 1,833,393** — the flat model overstates value by **USD 7,573** (0.4 %).
5. **Interpretation.** The flat 6 % "averages" the curve but misprices both flows — too harsh on the near one, too kind on the far one — and the errors do not cancel unless the stream is symmetric around the average tenor. For a two-flow example the drift is small; on a 25-year concession against a steep curve it moves tens of millions. The audit question is never "is the rate reasonable?" but "which curve does this rate summarise, and has the summary been stress-tested?"

3.A.4 The reviewer's TVM eye

Experienced reviewers do not recompute everything; they run invariants. The domain's collected set: factors decline; $DF(1)(1+r) = 1$; annuity factor × $r \to 1 - DF(n)$; schedules zero out; Σ principal = P; effective ≥ nominal, with equality only at annual compounding; real < nominal whenever $\pi > 0$; forward off-spot in the direction of the interest differential. Any violated invariant is a defect somewhere — find it before anyone downstream builds on it.

Industry variations — Domain 3

The machinery is universal; its parameters are sectoral, and a project finance leader reads the sector before trusting any inherited rate, tenor or convention:

- **Power and renewables.** Long contracted tenors (20–35 years) make value acutely curve-sensitive (3.A.3) and escalation-sensitive (3.3.2); availability and capacity payments are near-annuities, so the annuity family does most of the valuation work. Debt sculpting to seasonal cash flows (Domain 10) starts from monthly, not annual, discounting.
- **Transport concessions.** Patronage risk pushes discount rates up and makes single-rate valuation least defensible — scenario-based rates and revenue stress tests (Domain 7, KA 7.4) ride directly on this domain's factors.

- **Water and regulated utilities.** Regulatory reset cycles (often 5-yearly) partition the horizon: within-period flows discount at financing rates; across resets, the real/nominal discipline (3.3.1) dominates because regulators commonly work in real terms while lenders live in nominal.
- **Digital infrastructure.** Shorter economic lives and refresh capex compress tenors: bullet and mini-perm shapes (3.2.2) with deliberate refinancing (Domain 15) are standard, so the balloon-risk arithmetic of Case B is daily practice, not a cautionary tale.
- **Oil, gas and mining.** Commodity-linked revenue makes escalation assumptions (3.3.2) the loudest value driver, and multi-currency operations make the day-count and FX conventions (3.3.3–3.3.4) contractual battlegrounds rather than back-office detail.
- **Social infrastructure PPPs.** Availability payments in advance are common — the annuity-due reading (WE 3.2.1b) is the difference between winning and mispricing a bid.

Case study — Domain 3: choosing between an upfront grant and an availability stream (water / PPP)

Situation. Kestrel Water SPC's grantor offers two support packages for the desalination concession, and the board must choose one at financial close:

- **Package U:** a single upfront capital grant of **USD 58,000,000** at completion.
- **Package S:** an availability supplement of **USD 5,600,000 per year for 25 years**, paid annually in arrears from completion.

Kestrel evaluates support at its 8.0 % project discount rate. The grantor's advisers privately evaluate at 10 %.

Analysis. The stream's value to Kestrel: `AF(0.08, 25) = (1 − 1.08⁻²⁵)/0.08 = 10.674776`; `PV = 5,600,000 × 10.674776 =` **USD 59,778,747** — worth **USD 1,778,747 more** than the upfront grant. To the grantor's advisers at 10 %: `AF(0.10, 25) = 9.077040`; `PV = 5,600,000 × 9.077040 =` **USD 50,831,424** — worth **USD 7,168,576 less** than the grant they would otherwise pay today.

The decision. Both sides prefer a different package from the same facts — a discount-rate wedge, not a disagreement about arithmetic. Kestrel takes the stream (worth more at 8 %); the grantor is glad to give it (cheaper at 10 %); the deal closes with both counterparties better off by their own measure. The board minute records the rate, the convention (annual, in arrears) and the sensitivity: the stream's advantage to Kestrel disappears if its discount rate rises past ≈ 8.36 % — the **breakeven rate** at which `AF × 5,600,000 = 58,000,000`, i.e. `AF = 10.3571`.

What the domain teaches here. Valuation is always at a rate, for a party; streams and lumps are only comparable through `PV(x)`; and the negotiation surface between two parties' rates is where structuring value lives (Domain 7 revisits this as tariff design; Domain 9 as the grant-versus-support decision).

Case study B — Domain 3: the bullet that had to be refinanced (energy / refinancing)

Situation. A 30 MW peaking-plant SPV borrowed **USD 30,000,000** for 7 years. The sponsor chose a **bullet** (interest-only at 7.5 %, principal at maturity) over the lenders' preferred **annuity** shape, to keep early cash for distributions.

The two prices. Annuity instalment: $A = 30{,}000{,}000 \times 0.075/(1 - 1.075^{-7})$ = **USD 5,664,009** per year; total paid 39,648,066, of which interest **USD 9,648,066**. Bullet: interest $30{,}000{,}000 \times 0.075 =$ 2,250,000 per year — total interest **USD 15,750,000**, plus the full USD 30,000,000 due in year 7. The bullet pays **USD 6,101,934 more interest** for the privilege of deferral — and retains the entire principal as **refinancing risk**.

What happened. At year 7 the credit market had repriced: the refinancing cleared at 9.8 %, and the new 7-year annuity instalment on the rolled principal became $30{,}000{,}000 \times 0.098/(1 - 1.098^{-7}) = 6{,}121{,}646$ — against 5,664,009 had the original annuity simply been running to zero. Distributions locked up under the new facility's covenants for two years (Domain 10, KA 10.4).

What the domain teaches here. A schedule shape is a risk allocation, not a formatting choice: the annuity retires rate risk continuously; the bullet stores it at maturity, priced at a rate nobody today knows. Deferral has a computable minimum cost (the interest differential) and an uncomputable tail (the refinancing market) — leadership means pricing the first honestly and sizing the second deliberately (Domain 15, KA 15.3).

Executive perspective — Domain 3

What a project finance director cannot delegate in this domain:

- **The rate.** Analysts compute with a discount rate; directors own it. Whoever sets r sets the answer — the choice is a governance act (Domain 4 gives it method; Domain 3 makes its power visible).
- **The world.** Nominal or real, once, for the whole model — and the director asks which, out loud, at the first model review.
- **The conventions.** In-arrears vs in-advance, compounding frequency, day counts, escalation indices: each is small, each moves millions at scale, and together they are why the assumption register (3.T.1) is a board-visible artefact, not analyst hygiene.
- **The verification culture.** The invariants of 3.A.4 cost minutes; unverified TVM in a signed bid costs careers. The director's question is never "who computed this?" but "who recomputed this?" — and, where AI drafted it, the answer must name a human.

Calculation exercises — Domain 3

> **Exercise 3.1** A land payment of USD 850,000 falls due in 8 years; the applicable discount rate is 11 %. Value it today. Solution. $PV = 850{,}000 / 1.11^8 = 850{,}000 / 2.304538 =$ **USD 368,838** (368,837.52). Common error: using n = 7 gives 409,410 — an off-by-one from counting "in 8 years" as 7 compounding periods.

> **Exercise 3.2** A USD 2,500,000 equipment loan is repaid monthly over 20 years at a 5.0 % nominal annual rate compounded monthly. Find the monthly payment. Solution. Periodic rate $0.05/12 = 0.004167$; $n = 240$; $A = 2{,}500{,}000 \times 0.004167/(1 - 1.004167^{-240}) =$ **USD 16,499** per month (16,498.89). Common error: using the EAR (5.116 %) with 20 annual periods misstates debt service materially.

Exercise 3.3 Convert a 12 % nominal rate compounded monthly to its effective annual rate. Solution. `(1 + 0.01)¹² − 1 = 1.126825 − 1 =` **12.68 %**. Common error: reporting 12 % — the nominal — as "the rate" in a comparison against an annually-compounded 12.5 % offer, which is in fact cheaper.

Exercise 3.4 A transmission wayleave pays USD 1,200,000 per year indefinitely; the discount rate is 8.5 %. Value the perpetuity. Solution. `PV = 1,200,000 / 0.085 =` **USD 14,117,647**. Common error: applying a long annuity factor (say 30 years, `AF = 10.694`) "as approximately forever" understates value by ≈ 9 %.

Exercise 3.5 A market forecast promises a 12 % nominal return with 5 % inflation. State the real return, exactly and by the shortcut, and the shortcut's error. Solution. Exact: `1.12/1.05 − 1 =` **6.67 %** (6.6667 %). Shortcut: `12 − 5 = 7.00 %`. The shortcut overstates the real return by **33 basis points** — material over any long horizon.

Practitioner's toolkit — Domain 3

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 3.T.1 — TVM assumption register (one row per modelled stream)

Field	Discipline it enforces
Stream name & source document	Every number traces to a contract or estimate
Currency	Mismatches surface (KA 3.3.3)
Nominal or real	The consistency rule (3.3.1), stated not assumed
Rate & its owner	The director owns `r` (Executive perspective)
Compounding frequency & day count	Effective-rate honesty (3.2.3)
Timing convention (arrears/advance/mid-period)	Half-period errors caught (3.A.2)
Escalation index, lag, cap/floor	Contractual, not invented (3.3.2)

Toolkit 3.T.2 — Schedule QA checklist

- [ ] Payment recomputed independently of the model (different implementer or method).
- [ ] Final balance exactly zero; Σ principal = original loan.
- [ ] Periodic rate = nominal / frequency; period count = years × frequency.
- [ ] Discount factors monotone; `DF(1) × (1 + r) = 1`.
- [ ] Display rounding only — full precision beneath every printed figure.
- [ ] AI-drafted content marked, and its human verifier named (Domain 16, KA 16.4).

Toolkit 3.T.3 — Escalation register

For each escalating line: base amount and base date · index (publisher, series, definition) · lag · formula (full/partial indexation, weightings) · cap/floor · compounding basis · the clause reference in the contract. A model whose escalation register is complete can survive a Domain 13 model audit; one whose escalation is "4 % because last year" cannot.

Exam preparation — Domain 3

The calculation traps. Off-by-one periods ("in year 5" vs "after 5 years") · simple-for- compound substitution · rounded discount factors in the arithmetic · annuity-due priced as ordinary · EAR used as the periodic rate · nominal/real mixing · escalation applied simply · tenor mismatch in annuity factors (the 3.2-A distractor D) · treating a forward as a forecast.

Reflection questions. 1. Your CFO asks for "the value of the concession". What three questions must you answer before any number is defensible? (At what rate; in which world — nominal or real; under which timing conventions.) 2. A lender and a sponsor disagree about the value of the same payment stream. Under what conditions are both right — and where does that create negotiating room? (Case study A.) 3. Which invariant checks would have caught the last TVM error you saw in the wild — and why didn't they run? (3.A.4; toolkit 3.T.2.)

Domain 3 summary

Money's value is a function of time, and this domain built the complete conversion machinery: compounding and discounting (`FV(x)`, `PV(x)`, `DF(t)`); the annuity family that prices every level stream from availability payments to debt service (`AF(r,n)`, perpetuities, the three loan shapes and their risk meanings); frequency and effective-rate honesty; and the three real-world adjustments — Fisher-consistent inflation treatment, contract-grade escalation, and no-arbitrage currency forwards. Its discipline is as important as its formulae: one world per model, full precision beneath displayed rounding, conventions in the assumption register, invariants run on every schedule, and machine-drafted arithmetic verified by a named human before anyone relies on it. Every discounted number in the thirteen domains that follow stands on this one.

04

DOMAIN 4

Investment Appraisal and Capital Budgeting (quantitative)

Group: Foundations (Domain 4 of 4 in Part One). **Target:** ~70 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (`docs/books/registries/`). This domain is the home of the appraisal symbols — `NPV`, `IRR`, `MIRR`, `PI`, `EAV` — and builds directly on Domain 3's machinery (`PV(x)`, `DF(t)`, `AF(r, n)`): every formula here is a disciplined arrangement of those parts. British English; USD (+SAR where useful, indicative `USD 1 ≈ SAR 3.75`).

Why this domain exists

Domain 3 taught how to move money across time; this domain teaches how to decide with it. Investment appraisal is where analysis meets commitment: the techniques here — net present value, internal rates of return, payback, profitability, equivalent annual value — are how a project finance leader turns forecast cash flows into a defensible yes, no, or not-this-one. The measures are deliberately plural because each answers a different question: NPV measures value created; IRR measures the return embedded in the flows; payback measures exposure; the profitability index measures value per scarce dollar; equivalent annual value compares unlike lifetimes. Leadership in this domain means knowing which question the decision actually asks — and refusing to let a single seductive percentage (usually IRR) answer all of them. Domain 6 industrialises these calculations in the financial model; Domain 10 replays them from the lender's side of the table.

Learning objectives. After this domain a candidate can: compute and interpret NPV and explain why it is the primary measure of value; compute IRR, recognise its pathologies (multiple roots, scale-blindness, reinvestment assumption) and know when it misleads; compute MIRR and state what it fixes and what it does not; apply payback and discounted payback as exposure measures, not value measures; rank with the profitability index under capital rationing; compare unequal lives with equivalent annual value; resolve NPV–IRR conflicts on mutually exclusive projects; and subject any machine-produced appraisal to the family's verification rule.

The master appraisal. The Kestrel Water SPC financing of Domain 3 now faces its investment decision. Building the plant requires **I_0 = USD 60,000,000** today; the operating model forecasts **net cash inflows of USD 8,900,000 per year for 15 years**; the board's discount rate remains **8.0 %**. Every KA below interrogates this one decision from a different angle.

KNOWLEDGE AREA 4.1

The discounted measures: NPV, IRR and MIRR

Topics: 4.1.1 net present value · 4.1.2 IRR and its pathologies · 4.1.3 MIRR.

4.1.1 Net present value

Definition. NPV discounts every cash flow to today and nets off the investment:

$$NPV = \Sigma\, CF_t / (1 + r)^t - I_0$$

A positive NPV means the project returns more than the capital's required return r — it creates value; a negative NPV destroys it. NPV's authority rests on three properties no rival measure shares: it is **additive** (portfolio NPV is the sum of project NPVs), it is **scale-aware** (twice the project, twice the NPV), and it embeds the **correct reinvestment assumption** (cash thrown off is assumed to earn r, the opportunity cost — not the project's own return).

WORKED EXAMPLE

Worked example 4.1.1 — should Kestrel build?

1. **Setup.** I_0 = USD 60,000,000; net inflows USD 8,900,000 per year for 15 years (in arrears); r = 8.0 %.
2. **Formula.** For a level stream, $NPV = A \times AF(r, n) - I_0$ (Domain 3, KA 3.2).
3. **Substitution.** $AF(0.08, 15) = 8.559479$; $NPV = 8{,}900{,}000 \times 8.559479 - 60{,}000{,}000 = 76{,}179{,}360 - 60{,}000{,}000$.
4. **Result. NPV = +USD 16,179,360** (≈ SAR 60.7 million indicatively).
5. **Interpretation.** At the board's 8 % the plant is worth USD 16.2 million more than it costs — build, on these forecasts. The phrase carrying the professional weight is on these forecasts: NPV is only as honest as the cash flows and the rate beneath it, which is why Domain 6's model discipline and this domain's sensitivity habits (KA 4.3.3) are part of the appraisal, not decoration around it.

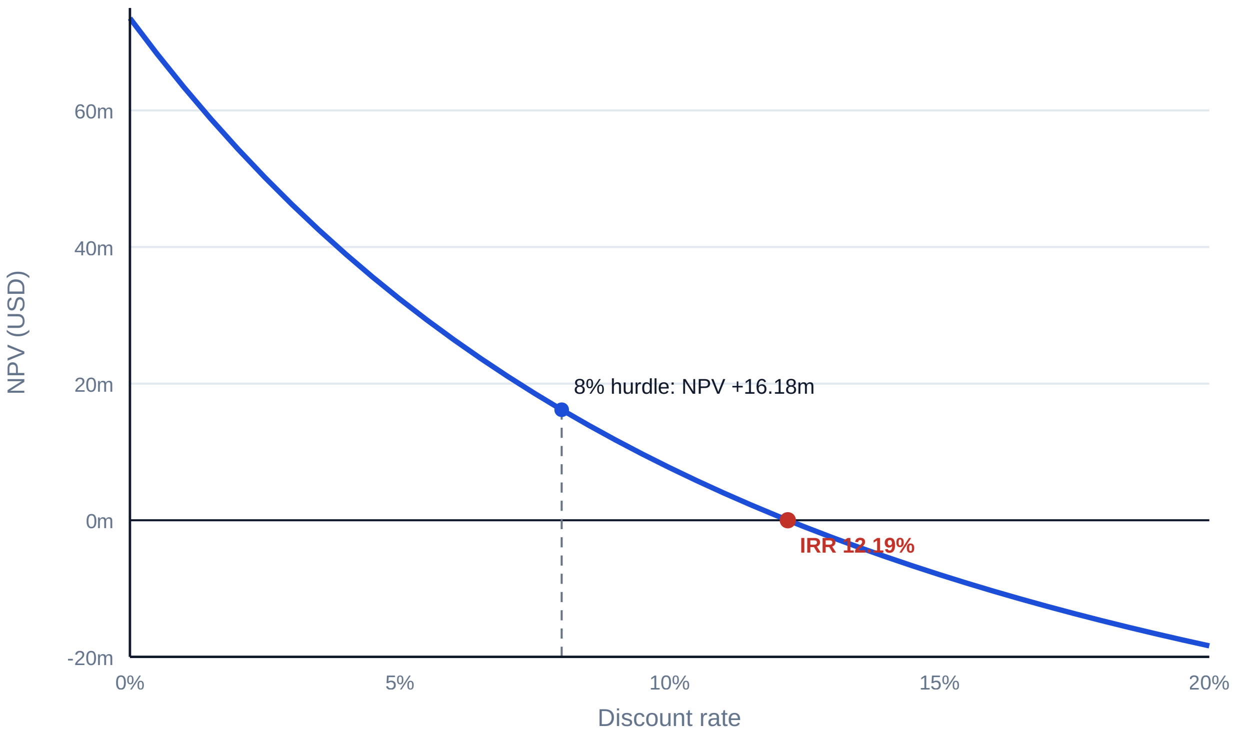

Fig 4.1.1 — Kestrel's NPV profile

4.1.2 IRR and its pathologies

Definition. The internal rate of return is the discount rate at which NPV is zero — the break-even price of capital:

```
NPV(IRR) = 0
```

For Kestrel: solve `8,900,000 × AF(r, 15) = 60,000,000` → `AF = 6.741573` → **IRR = 12.19 %**. Read against an 8 % hurdle, the project clears with 4.19 points to spare — the same verdict as NPV, expressed as a margin of safety rather than a value.

Why decision-makers love it — and where it lies. A percentage needs no context, which is precisely its danger. The three standing pathologies:

1. **Multiple roots.** IRR is a polynomial root; flows that change sign more than once can have more than one. The classic mining/decommissioning shape — invest, harvest, pay to restore — `(−1,000,000, +2,300,000, −1,320,000)` — satisfies NPV = 0 at **both 10 % and 20 %**; between them NPV is positive (about +1,890 at 15 %), outside, negative. Quoting "the IRR" of such a project is meaningless; NPV at the actual cost of capital remains well defined.
2. **Scale-blindness.** A 40 % IRR on USD 100,000 creates less value than 15 % on USD 20,000,000. IRR ranks intensity, NPV ranks money; mutually exclusive choices need money (KA 4.3.1).
3. **The reinvestment fiction.** IRR mathematically assumes interim cash is reinvested at the IRR itself — flattering for high-IRR projects, since no treasury desk reinvests at 28 %. MIRR (4.1.3) exists to repair exactly this.

Common pitfall — comparing IRRs across different tenors. A 3-year 18 % and a 15-year 13 % are not ordered: the short project leaves twelve years of reinvestment risk the percentage hides. The honest comparison is NPV over a common horizon, or EAV (KA 4.2.3).

4.1.3 MIRR — the modified internal rate of return

Definition. MIRR discounts outflows at the finance rate and compounds inflows to the terminal date at an explicit reinvestment rate, then reads the single rate connecting the two:

```
MIRR = ( FV(inflows at r_reinvest) / PV(outflows at r_finance) )^(1/n) − 1
```

WORKED EXAMPLE

Worked example 4.1.3 — Kestrel's honest percentage.

1. **Setup.** The master appraisal, with both finance and reinvestment rates set to the 8 % opportunity cost.
2. **Formula.** Terminal value of the inflows `TV = A × FVAF(r, n)` where $\mathrm{FVAF}(0.08, 15) = (1.08^{15} - 1)/0.08 = 27.152114$; then $\mathrm{MIRR} = (\mathrm{TV}/I_0)^{(1/15)} - 1$.
3. **Substitution.** `TV = 8,900,000 × 27.152114 = 241,653,814`; `MIRR = (241,653,814 / 60,000,000)^(1/15) − 1 = 4.027564^(1/15) − 1`.
4. **Result. MIRR = 9.73 %** — against an IRR of 12.19 %.
5. **Interpretation.** Two and a half points of Kestrel's IRR were the reinvestment fiction. MIRR is single-valued even for sign-changing flows and states its reinvestment assumption in the open — which is why credit committees increasingly ask for it alongside IRR. It remains a rate: still scale-blind, still not additive. The order of authority stands: **NPV decides; rates explain.**

AI in this KA

Appraisal is where optimisation pressure meets a suggestible tool: an assistant asked to "get the IRR above 12 %" will find assumptions that do. The governed workflow inverts the prompt — AI is asked to attack the appraisal (which assumptions move NPV most? where does the IRR become multiple?), the analyst reruns the golden checks (NPV at the stated rate recomputed independently; IRR verified by substitution back into the NPV equation; MIRR's terminal value re-derived), and the professional owns the recommendation. **AI proposes; the professional verifies, decides and remains accountable.**

KEY TERMS — KA 4.1

Term	Meaning
`NPV`	Σ discounted cash flows − investment; the primary value measure.
`IRR`	The rate at which NPV = 0; the flows' embedded return.
NPV profile	NPV plotted against the discount rate; IRR is its zero-crossing.
Multiple roots	Two or more IRRs when flow signs change more than once.
Reinvestment assumption	NPV assumes interim cash earns `r`; IRR assumes it earns the IRR.
`MIRR`	Single rate from explicit finance and reinvestment rates.

SAMPLE MCQS — KA 4.1

MCQ 4.1-A `[4.1.1 · Application]` A project costs USD 60,000,000 and returns USD 8,900,000 per year for 15 years; the discount rate is 8 % (`AF = 8.559479`). Its NPV is: - A. +USD 16,179,360 - B. +USD 73,500,000 - C. −USD 16,179,360 - D. +USD 76,179,360

Rationale: `8,900,000 × 8.559479 − 60,000,000 = +16,179,360`. B is the undiscounted surplus (8.9 × 15 − 60); C reverses the sign (investment minus value); D forgets to deduct the investment at all.

MCQ 4.1-B `[4.1.2 · Analysis]` A project's cash flows are −1,000,000, +2,300,000, −1,320,000. Which statement is correct? - A. its IRR is 10 % - B. its IRR is 20 % - C. it has two IRRs (10 % and 20 %), so decision by IRR is indeterminate and NPV at the cost of capital must decide - D. it has no IRR, so it must be rejected

Rationale: Two sign changes admit two roots — both 10 % and 20 % zero the NPV, which is positive between them. A and B are each half the truth and therefore wrong as "the" IRR; D confuses indeterminate ranking with non-viability — at a 15 % cost of capital the NPV is (slightly) positive.

MCQ 4.1-C `[4.1.3 · Application]` Using an 8 % finance and reinvestment rate, the master appraisal's terminal value of inflows is USD 241,653,814 on an investment of USD 60,000,000 over 15 years. MIRR is closest to: - A. 12.19 % - B. 9.73 % - C. 8.00 % - D. 26.85 %

Rationale: `(241,653,814/60,000,000)^(1/15) − 1 = 9.73 %`. A is the unmodified IRR; C is the reinvestment rate itself; D divides the 4.03× money multiple by 15 as if returns were simple.

MCQ 4.1-D `[4.1.1 · Analysis]` Which property justifies NPV's primacy over IRR for accept/reject and ranking decisions? - A. NPV is easier to compute - B. NPV is expressed as a percentage - C. NPV is additive and scale-aware, and assumes reinvestment at the opportunity cost - D. NPV never requires a forecast

Rationale: The three structural properties of 4.1.1. A is irrelevant and untrue at scale; B describes IRR, not NPV; D is false — NPV consumes the same forecasts as every measure.

SELF-CHECK — KA 4.1

1. State the master appraisal's three verdicts and their meanings. — NPV +16.18m (value created at 8 %); IRR 12.19 % (break-even rate, 4.19-point margin); MIRR 9.73 % (return with honest 8 % reinvestment).
2. Why does a mining-with-restoration cash flow break IRR? — Two sign changes → up to two roots; "the" IRR does not exist.
3. What single question decides between NPV and IRR when they disagree? — Which project adds more money at the actual cost of capital — NPV's answer, by construction.

KNOWLEDGE AREA 4.2

The complementary measures

Topics: 4.2.1 payback and discounted payback · 4.2.2 profitability index · 4.2.3 equivalent annual value.

4.2.1 Payback and discounted payback

Definitions. Payback is the time until cumulative cash inflows repay the investment; **discounted payback** repeats the question in present-value terms. Neither measures value — both measure **exposure**: how long the capital is at risk before it has come home.

WORKED EXAMPLE

Worked example 4.2.1 — how long is Kestrel's money in the ground?

1. **Setup.** The master appraisal: I_0 = 60,000,000; inflows 8,900,000 per year; r = 8 %.
2. **Formula.** Payback = I_0 / A for a level stream. Discounted payback: the smallest n with $A \times AF(r, n) \geq I_0$, interpolated within the crossing year.
3. **Substitution.** Simple: `60,000,000 / 8,900,000 = 6.74`. Discounted: after year 10 the cumulative PV is `8,900,000 × 6.710081 = 59,719,724` — USD 280,276 short; year 11's discounted flow is `8,900,000 × 0.428883 = 3,817,057`; fraction `280,276 / 3,817,057 = 0.07`.
4. **Result.** Payback **6.74 years**; discounted payback **10.07 years**.
5. **Interpretation.** The gap between the two numbers — three and a third years — is the price of time value: nominal repayment by year 7 does not return the capital's worth until year 10. For a 15-year concession, ten years of exposure is a risk fact the board should see beside the +16.2m NPV, not instead of it. The standing rule: **payback screens; NPV decides.** Payback's known biases — it ignores everything after the cut-off and penalises long-life infrastructure precisely for being long-lived — make it a veto-check, never a ranking.

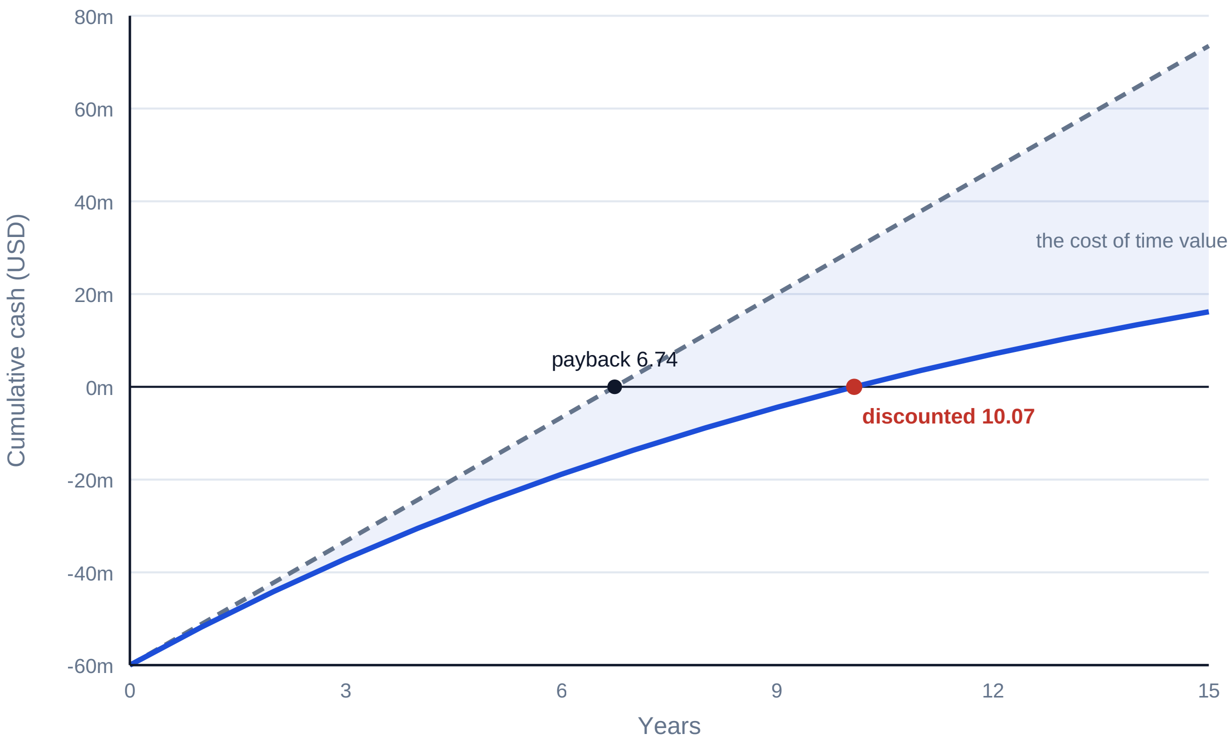

Fig 4.2.1 — Two paybacks: cumulative cash versus cumulative present value

4.2.2 The profitability index

Definition. Value created per unit of scarce capital:

```
PI = PV(future cash flows) / I₀
```

Kestrel: `76,179,360 / 60,000,000 = **1.270**` — each invested dollar buys USD 1.27 of present value. PI ranks efficiency and earns its keep in exactly one situation: **capital rationing** (KA 4.3.2), where the question is not "is this project good?" but "which portfolio of good projects fits the budget?". Unconstrained, PI adds nothing to NPV and can mis-rank mutually exclusive projects of different scale — the same scale-blindness as IRR, in ratio form.

4.2.3 Equivalent annual value

Definition. EAV converts an NPV over a life of `n` years into the level annual amount with the same present value:

```
EAV = NPV / AF(r, n)
```

Its natural habitat is **unequal lives**: comparing a 3-year asset with a 5-year asset by raw NPV smuggles in the assumption that the world ends when the shorter one does.

WORKED EXAMPLE

Worked example 4.2.3 — two pumps, unequal lives.

1. **Setup.** Kestrel must choose a dosing-pump system. **System A**: costs 5,000,000, runs 3 years, operating cost 800,000 per year. **System B**: costs 7,600,000, runs 5 years, operating cost 500,000 per year. Same duty; `r` = 8 %. (Costs, so lower is better.)
2. **Formula.** PV of cost = capex + opex × `AF(r, n)`; equivalent annual cost `EAC = PV / AF(r, n)`.
3. **Substitution.** A: `5,000,000 + 800,000 × 2.577097 = 7,061,678`; `EAC = 7,061,678 / 2.577097`. B: `7,600,000 + 500,000 × 3.992710 = 9,596,355`; `EAC = 9,596,355 / 3.992710`.
4. **Result.** A: **USD 2,740,168 per year**. B: **USD 2,403,469 per year** — B is cheaper by USD 336,699 every year, despite the 52 % higher purchase price.
5. **Interpretation.** Raw PV comparison (7.06m vs 9.60m) points the wrong way because it prices three years against five. EAV puts both on a per-year footing under the standard assumption that each system is replaced in kind at the end of its life. When that assumption fails — technology shifts, the duty ends — model the actual replacement chain instead (Domain 8's lifecycle costing).

AI in this KA

Screening portfolios is high-volume, formulaic work — the natural first place an organisation automates appraisal, and the first place systematic error industrialises. Two governed habits: the screening tool's formulae are validated once against this domain's golden examples (the registry pattern, `_build/verify_formulas.py`), and every auto-screened rejection above a materiality line gets a human review — a portfolio can lose its best project to one wrong sign convention, silently, forever.

KEY TERMS — KA 4.2

Term	Meaning
Payback / discounted payback	Years for cumulative (discounted) inflows to repay I_0; exposure measures.
PI	PV of inflows / I_0; value per scarce dollar, for rationing.
EAV / **EAC**	NPV (cost PV) converted to a level annual equivalent via AF(r, n).
Unequal lives	The comparison EAV exists to make honest.
Replacement chain	The explicit alternative when like-for-like replacement fails.

SAMPLE MCQS — KA 4.2

MCQ 4.2-A [4.2.1 · Application] For the master appraisal (60m in, 8.9m/yr, 8 %), the discounted payback is closest to: - A. 6.74 years - B. 10.07 years - C. 8.00 years - D. 15.00 years

Rationale: Cumulative PV reaches 59.72m after year 10; the 0.28m shortfall is 7 % of year 11's 3.82m discounted flow → 10.07. A is the simple payback; C confuses the discount rate with a duration; D is the whole life.

MCQ 4.2-B [4.2.3 · Application] System A (PV of costs 7,061,678 over 3 years, AF = 2.577097) versus System B (PV 9,596,355 over 5 years, AF = 3.992710). The correct comparison and choice is: - A. raw PV: A is cheaper, choose A - B. equivalent annual cost: A 2,740,168 vs B 2,403,469 — choose B - C. equivalent annual cost: A 2,353,893 vs B 1,919,271 — choose B - D. purchase price: A is cheaper, choose A

Rationale: Unequal lives require EAC: 7,061,678/2.577097 = 2,740,168 vs 9,596,355/3.992710 = 2,403,469. A and D compare unlike horizons; C divides each PV by its raw life in years (÷3 and ÷5), annualising without discounting — the right instinct with the wrong arithmetic.

MCQ 4.2-C [4.2.1 · Analysis] A board adopts "payback under 5 years" as its sole investment criterion. The predictable portfolio distortion is: - A. none — payback is conservative, so the portfolio is safe - B. systematic bias against long-lived infrastructure and toward short-cycle projects, regardless of value created - C. excessive investment in high-NPV projects - D. elimination of all risk

Rationale: Payback ignores everything beyond its cut-off, so a 15-year concession with NPV +16m loses to a 4-year project with NPV +1m. That is a value distortion, not conservatism (A); C reverses the effect; D confuses shorter exposure with no risk.

MCQ 4.2-D [4.2.2 · Recall] The profitability index earns its place in appraisal when: - A. projects are mutually exclusive and differ in scale - B. capital is rationed and the question is which portfolio of positive-NPV projects fits the budget - C. cash flows change sign more than once - D. lives are unequal

Rationale: PI ranks value per scarce dollar — the rationing question exactly. A is where PI (like IRR) mis-ranks by scale; C is MIRR's territory; D is EAV's.

SELF-CHECK — KA 4.2

1. Why is discounted payback always later than simple payback? — Discounting shrinks every inflow, so the cumulative line climbs more slowly (equal only if $r = 0$).
2. State Kestrel's PI and its meaning. — 1.270: each dollar invested buys 1.27 dollars of present value.

3. What assumption does EAV smuggle in, and when must you drop it? — Like-for-like replacement in perpetuity; drop it when the duty ends or technology shifts, and model the real chain.

KNOWLEDGE AREA 4.3

Decision contexts: exclusivity, rationing and judgment

Topics: 4.3.1 mutually exclusive investments · 4.3.2 capital rationing · 4.3.3 the limits of the numbers.

4.3.1 Mutually exclusive investments

The conflict. Choosing one project instead of another is where NPV and IRR famously disagree. Kestrel's board weighs two intake designs (same risk class, r = 8 %):

Design	I_0	Net inflow × 5 yrs	NPV @ 8 %	IRR
P (compact)	5,000,000	2,000,000	**+2,985,420**	**28.65 %**
Q (full-scale)	20,000,000	6,000,000	**+3,956,260**	**15.24 %**

IRR shouts P; NPV says Q adds USD 970,840 more money. The tie-breaker is the **incremental project** Q−P: invest a further 15,000,000 for a further 4,000,000 per year — an incremental IRR of **10.42 %**, above the 8 % cost of capital. The extra capital earns its keep, so take the bigger project. **Rule: when exclusivity forces a choice, rank by NPV; use incremental IRR only to narrate the same answer.** (At any hurdle above 10.42 % the ranking genuinely flips — the crossover rate is where the two NPV profiles intersect.)

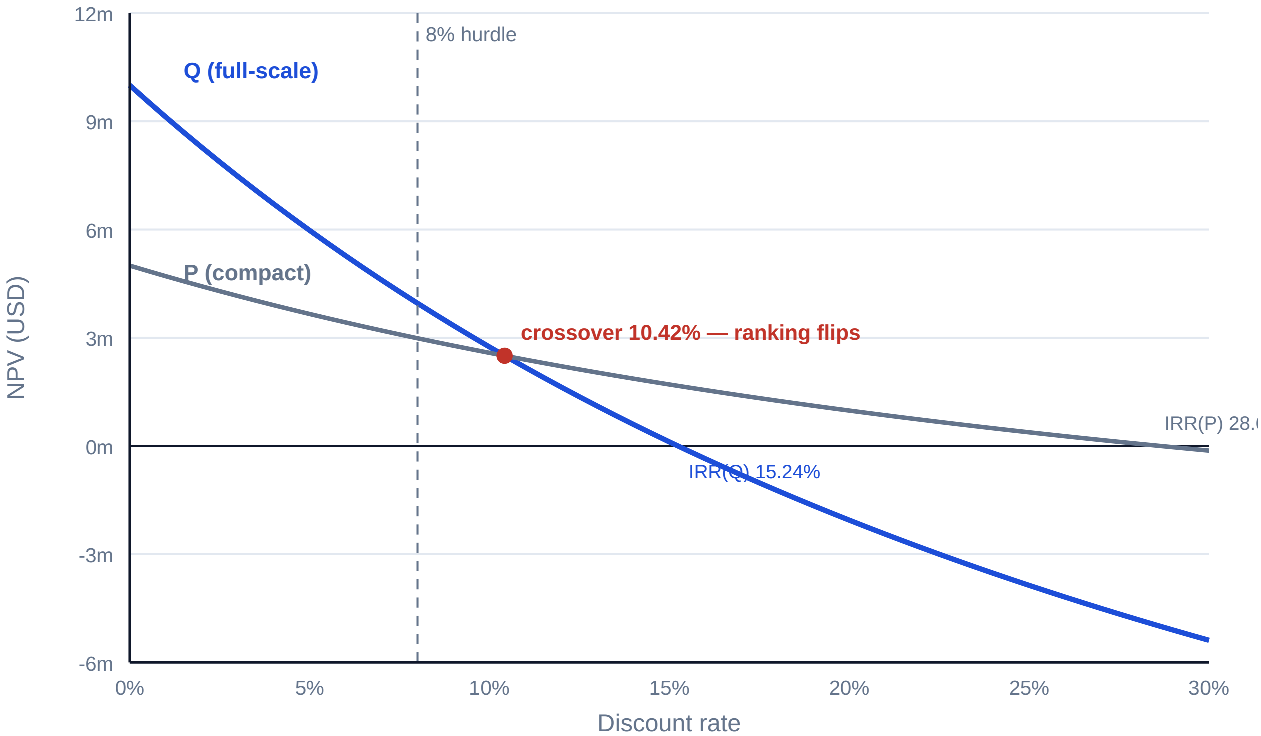

Fig 4.3.1 — Two NPV profiles and the crossover

4.3.2 Capital rationing

The problem. When the budget cannot fund every positive-NPV project, maximise **NPV per budget dollar** — rank by PI and pack the budget:

Project	I_0 (m)	NPV (m)	PI	Cumulative I_0
W	8	2.40	1.300	8
X	12	3.00	1.250	20 ← budget
Y	10	2.20	1.220	—
Z	6	1.02	1.170	—

With a USD 20m budget the PI ranking funds **W + X for NPV +5.40m**; the tempting "biggest NPV first, then fill" (X, then Y won't fit, then W) lands on the same 5.40m here, but the next-best feasible set (W + Y, +4.60m) shows what a wrong packing costs. Discipline points: PI packing is exact only when projects are divisible or the budget binds once — lumpy multi-period rationing is an optimisation problem (Domain 6's model does it honestly); and rationing itself deserves challenge, because turning away a 1.17-PI project is a financing failure as often as a screening success (Domain 9).

4.3.3 The limits of the numbers

Every measure in this domain consumes forecasts and a rate, and both are judgments wearing decimals. The professional frame: **appraisal quantifies; it does not decide strategy.** Numbers cannot see option value (the pilot that buys the right to scale), strategic foreclosure (the market position lost by not building), or the asymmetry of forecast error (Domain 8's ranges belong beside every point NPV). The board paper this domain endorses shows: NPV at the owned rate; the sensitivity of that NPV to the two or three assumptions that dominate it; IRR/MIRR as narrative; exposure via discounted payback; and the explicit statement of what the numbers cannot price. Anything less is theatre with spreadsheets.

AI in this KA

Ranking under constraints is combinatorial — machine territory — and constraint errors are silent. The governed pattern from Domain 6 applies unchanged: the optimiser proposes the portfolio; the analyst verifies the constraint set against reality (is the budget truly annual? are W and X really independent?); golden checks re-verify the NPVs being packed; and the leader owns the strategic overrides the model cannot see (4.3.3) — recorded as decisions, not adjustments smuggled into assumptions.

KEY TERMS — KA 4.3

Term	Meaning
Mutually exclusive	Choosing one forecloses the other; rank by NPV.
Incremental IRR	IRR of the difference project; narrates the NPV ranking.
Crossover rate	Rate where two NPV profiles intersect and rankings flip.
Capital rationing	Budget binds before value runs out; pack by PI.
Option value	Value of flexibility the static NPV cannot see.

SAMPLE MCQS — KA 4.3

MCQ 4.3-A `[4.3.1 · Application]` P: NPV +2,985,420, IRR 28.65 %. Q: NPV +3,956,260, IRR 15.24 %. Cost of capital 8 %; only one may proceed. The correct choice is: - A. P — higher IRR - B. Q — higher NPV, confirmed by an incremental IRR of 10.42 % above the hurdle - C. P — lower investment is always safer - D. both, split the budget

Rationale: Exclusive choices rank by money added: Q adds USD 970,840 more, and the extra 15m earns 10.42 % > 8 %. A is the scale-blindness pathology; C prices fear, not value; D violates the premise.

MCQ 4.3-B `[4.3.2 · Application]` Budget USD 20m. Projects (I_0, NPV): W (8, 2.4), X (12, 3.0), Y (10, 2.2), Z (6, 1.02). The value-maximising funded set is: - A. X and Y - B. W and X: NPV +5.40m - C. W, Y and Z - D. X and Z

Rationale: PI order W (1.300), X (1.250) packs the budget exactly for +5.40m. A needs 22m; C needs 24m; D fits (18m) but yields +4.02m — a 1.38m sacrifice for leaving W unfunded.

MCQ 4.3-C `[4.3.3 · Analysis]` A pilot plant shows NPV −800,000, but building it creates the option to deploy at 20× scale if the technology proves. The appraisal-sound treatment is: - A. reject — negative NPV is disqualifying - B. approve by overriding the NPV silently - C. present the static NPV alongside an explicit valuation (or structured judgment) of the scaling option, and decide on the combined case, recorded as such - D. raise the forecast cash flows until NPV turns positive

Rationale: Static NPV cannot see option value; the remedy is to price or judge the option in the open. A discards real value; B and D are the same dishonesty at different altitudes — one hides the judgment, the other disguises it as a forecast.

MCQ 4.3-D `[4.3.1 · Recall]` The crossover rate of two NPV profiles is: - A. the rate at which both projects' NPVs equal zero - B. the rate at which the two NPVs are equal — above it, the ranking flips - C. the average of the two IRRs - D. the cost of capital

Rationale: Crossover is the intersection of profiles (equivalently the incremental project's IRR). A describes two separate IRRs; C is arithmetic superstition; D is a property of the firm, not of the pair.

SELF-CHECK — KA 4.3

1. Why does incremental IRR always agree with NPV on a pairwise choice? — It tests whether the difference project clears the hurdle, which is exactly what the NPV difference measures.
2. When does PI packing fail as a rationing rule? — Lumpy projects and multi-period budget constraints; then it's an optimisation, not a sort.
3. Name three things a positive NPV cannot tell the board. — Exposure duration; sensitivity concentration; the value of flexibility and strategic position outside the modelled flows.

Advanced topics — Domain 4

4.A.1 Reading NPV profiles like a professional

The profile (Fig 4.1.1) compresses the whole appraisal into one curve: its height at 0 % is the undiscounted surplus; its slope measures duration-sensitivity (long-dated flows steepen it); its zero-crossing is the IRR; and the distance between the hurdle and the crossing is the margin for rate error. Two profiles per decision (Fig 4.3.1) add the crossover — the complete geometry of a mutually exclusive choice. A reviewer who asks for the profile instead of the point estimate has converted an argument about numbers into an inspection of shape.

4.A.2 Inflation-consistent appraisal

The Fisher discipline of Domain 3 (KA 3.3.1) binds hardest here: nominal flows with the nominal rate, or real flows with the real rate — and the same world for every line, including the terminal value. The classic appraisal defect is a nominal hurdle (board-set, inflation-inclusive) discounting real flows (an engineer's "today's money" forecast): NPV is systematically understated and good projects die politely. The assumption register names the world; the reviewer checks one line item end to end.

4.A.3 The reviewer's appraisal eye

The invariants: NPV at 0 % equals the raw flow sum; NPV at the IRR equals zero (substitute it back — the cheapest IRR audit that exists); MIRR lies between the reinvestment rate and the IRR; PI > 1 ⇔ NPV > 0 at the same rate; EAV × `AF(r, n)` reproduces the NPV; incremental IRR above the hurdle ⇔ the bigger project's NPV is higher. Any violated line is a defect somewhere — the appraisal analogue of Domain 3's factor-table checks, and wired into this programme's golden-answer harness.

Industry variations — Domain 4

- **Regulated utilities and availability PPPs.** Contracted revenue narrows the forecast range, so appraisal battles concentrate on the rate (and the regulator's allowed return); NPV margins are thin and Fisher errors (4.A.2) are decision-changing.
- **Merchant power and commodities.** The forecast is the battlefield: point NPVs are close to meaningless without the scenario set (Domain 7's stress tests), and payback recovers status as a survival measure — how long must prices hold?
- **Oil, gas and mining.** Decommissioning liabilities put the sign-change pathology (4.1.2) on almost every appraisal: MIRR and NPV are standing practice, "the IRR" is treated with suspicion by convention.
- **Technology and corporate transformation.** Option value (4.3.3) dominates: staged investments are the norm, and static NPV is a floor, not an answer. Rationing is annual and lumpy — the optimisation caveat of 4.3.2 is the daily reality.
- **Public-sector appraisal.** The discount rate is policy (a social time preference rate, set centrally), distributional effects sit beside NPV, and appraisal guidance is published — the numbers travel with a governance file, which is the direction this domain pushes every sector.

Case study — Domain 4: the intake decision, taken properly (water / desalination)

Situation. Kestrel's board must (1) confirm the plant investment, (2) choose between intake designs P and Q, and (3) allocate the sponsor group's residual USD 20m development budget across four unrelated feeder projects (W–Z of KA 4.3.2). One meeting, three different appraisal questions.

The decisions. Build: NPV +16.18m at 8 %, IRR 12.19 %, MIRR 9.73 %, discounted payback 10.07 years — build, with the exposure noted and the two dominating assumptions (tariff escalation, availability) sensitivity-tested per KA 4.3.3. Intake: Q over P — +970,840 more NPV, incremental IRR 10.42 % over the extra 15m; the 28.65 % on P is recorded as the intensity it is, not the ranking it isn't. Budget: W + X for +5.40m, with the un-funded 1.17-PI project Z referred to Domain 9's financing question rather than silently killed.

What the domain teaches here. Three questions, three measures, one hierarchy: NPV decided all three calls; IRR, MIRR, PI and payback each explained a facet the board needed narrated. The minute records rates, worlds (nominal), sensitivity owners and the option-value judgments — the appraisal file is the governance file.

Case study B — Domain 4: the fund that bought percentages (infrastructure fund)

Situation. An infrastructure fund with a 9 % cost of capital repeatedly preferred short-tenor, high-IRR deals — the P-shape — over long-tenor concessions at 13–15 % IRR. Five years on, its realised portfolio return trailed its own reported deal IRRs by nearly four points.

What happened. The gap was the reinvestment fiction industrialised: each 25–30 % IRR assumed its distributions would earn the same, but the fund redeployed at market rates — when it could redeploy at all. MIRR at honest reinvestment rates (the 4.1.3 arithmetic) had flagged the true economics of every deal; it was computed, and filed. The long concessions it passed over — lower IRR, far higher NPV per dollar and decade — went to competitors who now hold them.

What the domain teaches here. A portfolio of intensities is not a portfolio of value. IRR's flattery compounds at fund level because short deals recycle the fiction; NPV and MIRR discipline is not academic tidiness — over a fund's life it is the difference between the return promised and the return delivered.

Executive perspective — Domain 4

What a project finance director cannot delegate in this domain:

- **The hurdle.** As in Domain 3: whoever sets r sets the answer. The director owns the rate, its world (nominal or real), and the standing sensitivity band around it.
- **The measure hierarchy.** NPV decides; rates and ratios explain. The director enforces this in every paper — and personally reads the profile, not the point.
- **The exclusivity discipline.** Incremental thinking (Q−P) is the director's question in the room: "what does the extra money earn?" beats "which percentage is bigger?" every time.
- **The honesty of rejections.** Under rationing, what was not funded and why is board information; a rejected 1.17-PI project is a financing gap wearing a screening verdict.
- **The unpriced.** Option value, strategic position, forecast asymmetry — the director's overrides are legitimate exactly when they are explicit, minuted and owned (4.3.3).

Calculation exercises — Domain 4

> **Exercise 4.1** I_0 = 12,000,000; net flows 3,000,000 · 4,000,000 · 5,000,000 · 5,000,000 in years 1–4; r = 10 %. Compute the NPV. Solution. PV = $3/1.10 + 4/1.10^2 + 5/1.10^3 + 5/1.10^4$ (millions) =

> 2,727,273 + 3,305,785 + 3,756,574 + 3,415,067 = 13,204,699; **NPV = +1,204,699**. Common error: discounting year 1 at `t` = 0 (no discount) shifts every factor one period and overstates NPV by ≈ 1.19m.

> **Exercise 4.2** I_0 = 9,000,000; level inflow 1,500,000 for 10 years. Find the IRR. Solution. `AF(r, 10) = 9/1.5 = 6.0`; solving gives **IRR = 10.56 %** (AF(10 %) = 6.145, AF(11 %) = 5.889 bracket it). Common error: reading the payback reciprocal (1.5/9 = 16.7 %) as the IRR — that shortcut holds only for perpetuities.

> **Exercise 4.3** I_0 = 10,000,000; inflows 2,600,000 for 6 years; reinvestment and finance rate 7 %. Compute MIRR. Solution. `FVAF(0.07, 6) = 7.153291`; TV = 18,598,556; `MIRR = (1.859856)^(1/6) − 1 =` **10.90 %**. Common error: compounding the inflows for 6 periods each (rather than to a common terminal date) inflates TV and the rate.

> **Exercise 4.4** Machine C1: 3,000,000 capex, 4-year life, 250,000/yr opex. Machine C2: 4,200,000, 6-year life, 150,000/yr. Same duty; `r` = 9 %. Choose. Solution. C1: PV = 3,000,000 + 250,000 × 3.239720 = 3,809,930; EAC = **1,176,006**. C2: PV = 4,200,000 + 150,000 × 4.485919 = 4,872,888; EAC = **1,086,263**. **Choose C2** — cheaper by 89,743 per year. Common error: comparing PVs across unequal lives (which picks C1).

> **Exercise 4.5** Budget 25,000,000. Projects (I_0; NPV): A (10; 2.60) · B (15; 3.30) · C (8; 1.84) · D (7; 1.40). Fund the best set. Solution. PIs: A 1.260 · C 1.230 · B 1.220 · D 1.200. **Greedy PI packing fails here**: A + C + D fits (25) for +5.84m, but **A + B (25 exactly) yields +5.90m**. With lumpy projects the rule is: use PI to shortlist, then enumerate feasible sets (or optimise). This is the 4.3.2 caveat as arithmetic.

Practitioner's toolkit — Domain 4

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 4.T.1 — The appraisal one-pager (per decision)

Decision question (accept/reject · exclusive choice · rationing) · cash-flow source and world (nominal/real, per 3.T.1) · rate and its owner · **NPV at the owned rate** · profile sketch or IRR/MIRR · discounted payback · PI (if rationing) · EAV (if unequal lives) · two-way sensitivity on the dominating assumptions · unpriced factors, stated · recommendation and the named accountable approver.

Toolkit 4.T.2 — IRR pathology checklist

- [ ] Sign changes counted; more than one → MIRR + NPV only, "the IRR" banned from the paper.
- [ ] Scale stated beside every percentage (I_0 and NPV in money).
- [ ] Tenors aligned before any cross-project rate comparison (else EAV).
- [ ] Reinvestment assumption named; MIRR computed when IRR > hurdle + 5 points.
- [ ] IRR substituted back: NPV(IRR) recomputes to zero.
- [ ] Incremental IRR computed for every exclusive pair; agrees with NPV ranking.

Toolkit 4.T.3 — Rationing worksheet

Columns: project · I_0 · NPV · PI · strategic notes/dependencies. Steps: rank by PI → shortlist to ~2× budget → enumerate feasible sets (lumpy) → record funded set, rejected set with reasons, and the financing referral for high-PI rejects (Domain 9).

Exam preparation — Domain 4

The calculation traps. Forgetting to deduct I_0 (MCQ 4.1-A distractor D) · discounting from $t = 0$ (Exercise 4.1) · payback reciprocal as IRR (Exercise 4.2) · ranking exclusives by IRR (4.3.1) · greedy PI with lumpy budgets (Exercise 4.5) · comparing PVs across unequal lives (Exercise 4.4) · quoting one IRR for sign-changing flows (4.1.2) · MIRR terminal value mis-compounded (Exercise 4.3) · nominal/real mixing imported from Domain 3.

Reflection questions. 1. Your board paper shows one number: "IRR 22 %". List the five questions this domain obliges you to ask before anyone votes. (Scale? Sign changes? Tenor? Reinvestment? NPV at our rate?) 2. When is a negative-NPV decision defensible — and what must the record contain? (4.3.3: explicit option/strategic value, owned and minuted.) 3. Which invariant of 4.A.3 would have caught the last appraisal error you saw? (Most often: NPV(IRR) ≠ 0 on substitution, or EAV × AF ≠ NPV.)

Domain 4 summary

Appraisal turns Domain 3's machinery into decisions, under one hierarchy: **NPV decides; everything else explains.** NPV earns primacy by additivity, scale-awareness and the honest reinvestment assumption; IRR narrates a margin of safety but carries three standing pathologies — multiple roots, scale-blindness, the reinvestment fiction — of which MIRR repairs only the last; payback and its discounted form measure exposure, never value; PI ranks value per scarce dollar under rationing (with the lumpy-budget caveat proven in Exercise 4.5); EAV makes unequal lives commensurable. Mutually exclusive choices resolve by NPV, narrated by incremental IRR and the crossover geometry; rationing resolves by disciplined packing plus the financing question for what was turned away; and the limits of the numbers — options, strategy, forecast asymmetry — are stated in the open, as owned judgments. The Kestrel thread continues into Domain 5 (is the project bankable, not merely valuable?) and Domain 6 (the model that industrialises every calculation this domain taught).

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